

The Foundation of Projection Relativity III

Zero-Parameter Radiative and Cross-Sector Closure of the Projection Ledger

Michael Stanislaus Oshetski 

michael.oshetski@micatu.com
Founder and Chief Technology Officer
Advanced Photonics Laboratory (APL)
Micatu, Inc.
Horseheads, New York, USA
ORCID: 0009-0007-3623-7586

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Abstract

Projection Relativity III (PR-III) studies the finite-tier radiative and cross-sector closure of the theoretical framework introduced in the PR-I and PR-II preprints. Building upon the frozen PR-I spectral-projection geometry and the PR-II compact electroweak and flavor lift, this work evaluates electromagnetic, electroweak, neutrino, strong, anomaly, and running-continuity branches without importing the selected diagnostic observables as fitted inputs. The result is a closed, reproducible, no-fit equation ledger at the stated finite tier.

Numerically, the generated architecture achieves sub-sigma agreement with empirical targets. The fine-structure inverse matches its reference to within $+0.1376\sigma$, while the core electroweak constants M_W , M_Z , and G_F align to within $+0.3566\sigma$, $+0.0364\sigma$, and $+0.6003\sigma$, respectively. Extending to the lepton sector, the neutrino branch is radiatively stable under the declared finite-tier multiplicative envelope and preserves the normal mass ordering, with the summed mass, effective beta observables, and $m_{\beta\beta}$ envelope remaining well below current empirical ceilings. The strong-sector ledger locks the $SU(3)_c$ color identities, active-flavor thresholds, and the one-loop asymptotic-freedom sign. This constrained architecture ultimately yields a strong anchor that agrees with diagnostic references to within 0.04%. Global structural audits successfully verify the cancellation of local gauge anomalies per generation and the absence of the Witten anomaly. Additionally, the residual electromagnetic branch remains purely vector-like, maintaining running continuity under strict no-fit provenance.

The charged-adjoint representation fixes the exact normalized quadratic invariant $\text{Tr}_{\text{ch}}(q^\dagger q) = 2$. PR-III identifies the neutral C_3 term with the minimal physical- Z response: a charged-preserving, photon-null rank-one correction built from the inherited scale ρ_{PR} and the unique additive, basis-invariant normalized quadratic charge functional. Within this declared admissible class the coefficient is uniquely $k = 2$, and no fitted freedom remains. Exact normal-form reduction shows that every other in-class equation either has nonzero defect $\rho_{\text{PR}}^2(k - 2)^2$ or is an invertible presentation of the same residual law on the complete declared charge-operator domain. The canonical finite-tier identity is established by the normalized-response theorem; a gauge-fixed neutral-Hessian presentation, when used, is an equivalent microscopic representation of the same in-class response law. Broader operator families with independent higher-dimensional Wilson coefficients define different theory classes and are not additional PR parameters or alternative PR branches. The reported C_3 value is therefore the canonical no-fit neutral closure at the stated tier, with its class-relative theorem status explicit. With the completed PR-IV Global Singleton Theorem, the same normal-form, exact-defect, and equivalence trichotomy closes the registered PR-I–PR-III equation architecture: its admissible quotient is a singleton with no fitted or free continuous parameter. This statement is relative to the declared postulates and finite-tier domain; additional fields, higher orders, and independent EFT coefficients define other theory classes rather than additional PR solutions. The numerical branch, software, and audit checks are available in the public repository.

1 Introduction

Projection Relativity III (PR-III) investigates the finite-tier radiative and cross-sector closure of the Projection Relativity framework. Rather than establishing a new theoretical starting point, this manuscript builds directly upon the geometric and topological foundations laid in the Projection Relativity I [9] and Projection Relativity II [10] preprints.

In PR-I, we introduced the master projection field $\Psi(x, \xi)$ and the minimal internal geometry

$$\mathcal{M}_{\text{int}} = \mathbb{R}_w \times S_\theta^1. \quad (1)$$

This architecture demonstrated that gravitational stiffness, displacement-generated inertia, and compact electromagnetic phase are not unrelated phenomena, but primary observable projections of a single internal spectral manifold. The radial spectral gap of this manifold supplied a finite coercive scale, enabling the finite-core regularization of the strong-field branch while preserving the exterior relativistic limit [9].

PR-II subsequently lifted this compact boundary construction into the non-Abelian electroweak and flavor setting [10]. By replacing the minimal phase circle with the parent compact electroweak fiber

$$\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L, \quad (2)$$

the residual electromagnetic phase was recovered as the locked boundary of the larger fiber [10]. This allowed weak mixing, the weak-scale structure, flavor hierarchy, and neutrino candidates to emerge as outputs of projection geometry rather than empirical insertions. The class-relative three-generation statement combines the broken-quotient dimension with a complete and nonredundant sheet-incidence map; rank-nullity then gives

$$N_{\text{gen}}^{\text{PR}} = \dim [(SU(2)_L \times U(1)_Y)/U(1)_{\text{em}}] = 3. \quad (3)$$

Dimension alone is not claimed to determine the physical sheet map. To make the full incidence equation unique under normalized basis changes, the normalized-incidence closure additionally requires $\Phi_3^\dagger \Phi_3 = I_3$; all such maps belong to one orthogonal equivalence orbit in the real realization (unitary after complexification).

The present paper begins from these frozen geometric structures. Its purpose is to close the declared finite-tier radiative, strong-sector, and running-continuity architecture of PR-I and PR-II. PR-III establishes reproducible consistency, stability, and no-fit provenance at that tier, and the companion PR-IV Global Singleton Theorem supplies equation-level exhaustion for the completed selector class. Together they define a closed class-relative architecture with no inequivalent equation satisfying the same admissibility rules remaining.

1.1 Structural Roadmap and Organization of the Paper

To audit this finite-tier cross-sector ledger, the remainder of the paper is organized as a sequential chain, outlined in Table 1. PR-I supplies the spectral-projection foundation. PR-II supplies the compact electroweak and flavor lift. PR-III proves stability and mutual consistency of the implemented finite-tier branches without importing target observables as fitted inputs. The theorem is closed in two layers: this manuscript establishes the declared finite-tier sectors, and PR-IV proves that their completed selector quotient contains one equivalence class.

1.2 Reader Prerequisites

A reader can follow the main logic of this paper by keeping several foundational concepts in view. From PR-I, this includes the master-field picture $\Psi(x, \xi)$ and the minimal internal geometry $\mathbb{R}_w \times S_\theta^1$, which establish

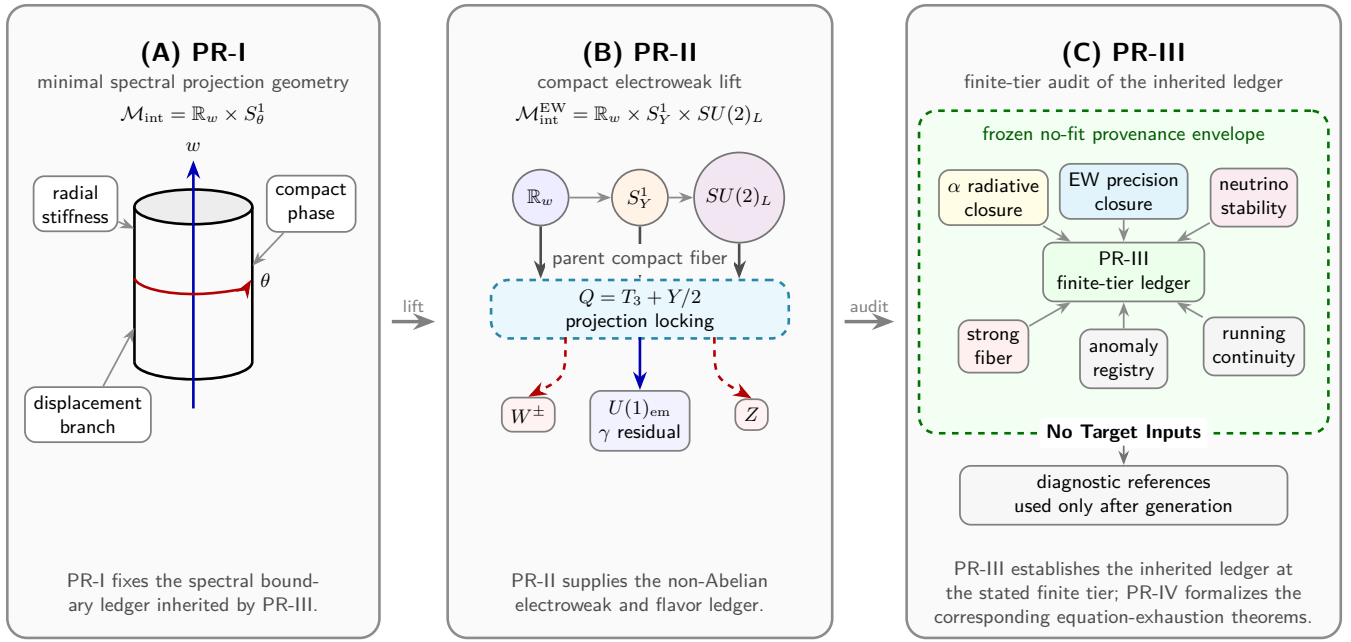


Figure 1: Geometric inheritance and no-fit closure across PR-I–PR-III. (A) PR-I begins with the minimal internal projection geometry $\mathcal{M}_{\text{int}} = \mathbb{R}_w \times S_\theta^1$, in which the non-compact radial direction supplies the stiffness branch and the compact phase direction supplies the residual electromagnetic phase. (B) PR-II lifts this compact boundary into the electroweak parent fiber $\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L$. Projection locking through $Q = T_3 + Y/2$ selects the unbroken residual $U(1)_{\text{em}}$ boundary, while the broken directions generate the massive $W^\pm$ and Z projection excitations. (C) PR-III does not introduce a new topology; it evaluates a declared finite-tier extension of the inherited PR-I/PR-II ledger inside a no-fit provenance envelope. The electromagnetic, electroweak, neutrino, strong, anomaly, running-continuity, and provenance branches are generated and audited without using their diagnostic reference values as generation inputs. The green enclosure denotes the closed finite-tier admissibility domain and its audit provenance; structures with additional orders, fields, coefficients, or postulates belong to a different theory class rather than to an alternative PR-III branch.

the interpretation of radial stiffness, displacement, and compact phase as unified projection sectors [9]. This foundation also provides the fixed PR-I boundary constants and the electromagnetic phase normalization [10]. From PR-II, the core prerequisites are the electroweak lift into the parent fiber $\mathbb{R}_w \times S_Y^1 \times SU(2)_L$, the recovery of the residual electromagnetic boundary via $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$, and the geometric generation-sheet rule $N_{\text{gen}}^{\text{PR}} = 3$ [10]. We note, the reader must keep in mind the strict methodological distinction between internally generated values and post-generation diagnostic references. The relevant prior derivations are reviewed here only where they are needed to define the PR-III ledger. Readers seeking the complete geometric construction should consult PR-I for the spectral-projection foundation and PR-II for the compact electroweak and flavor lift.

1.3 The No-Fit Provenance Discipline

A central rule is maintained throughout this manuscript: diagnostic references may evaluate PR-generated quantities, but they may never generate or tune them. Every value in this paper is strictly classified as either an inherited input from PR-I or PR-II, a PR-generated output, an explicitly declared closure quantity, or a post-generation diagnostic reference. This “no-fit” provenance discipline is why this manuscript distinguishes among structural identities, closure theorems, finite-tier relations, and diagnostic comparisons [9]. Absence of fitting is necessary for prediction, but it is logically distinct from the origin of a closure relation. A finite-tier relation may be admitted either by direct derivation from inherited PR operators or as an explicitly declared logical closure extension. In the latter case its admissible class, structural

Section	Sector / Phase	Finite-Tier Actions and Objectives
2 & 3	Foundation	Fixes the inherited PR-I/PR-II boundary ledger and alpha baseline policy, and reviews the scalar Hessian seed and gauge-null quotient needed to preserve the mass ledger.
4	Electromagnetic	Constructs the radiative extraction, locks the charged-mode supertrace inventory, and generates boundary-leakage corrections prior to comparison with external fine-structure references.
5	Electroweak	Transports the electromagnetic closure into the electroweak running scale, refining the weak-angle, vector self-energy, and neutral-channel ledgers to current diagnostic precision.
6	Neutrino	Audits the mass branch for radiative stability, normal ordering, and PMNS unitarity without importing external oscillation data or cosmological bounds as generation inputs.
7	Strong	Locks the compact color-fiber architecture, assembling group identities, active-flavor thresholds, and strong anchors without treating external QCD quantities as fitted targets.
8 & 9	Global Consistency & Uniqueness	Synthesizes the completed sectors into a global consistency statement, including the anomaly registry and cross-sector diagnostics, and states the closed admissibility domain and class-relative equation-exclusion theorem.

Table 1: The PR-III structural roadmap and paper organization, detailing the sequential closure of the inherited ledger.

motivation, normalization, and uniqueness must be stated, and no empirical target or fitted continuous coefficient may select it. Section 5.5 applies this standard to the neutral C_3 branch.

1.4 Reproducibility and Test-Driven Method

The development of PR-III follows the same test-driven standard as demonstrated in the previous manuscripts. Each construction is paired with a numerical or symbolic validation harness. The harnesses confirm the generated equations and include control cases that verify exclusion of structures incompatible with the declared PR ledger, including Abelianized $SU(2)$, photon-mass leakage, altered Fermi normalization, nonunitary flavor rotations, anomaly-incompatible fermion ledgers, inverted suppression factors, and target-observable leakage.

In PR-III this standard is applied to the full radiative and cross-sector ledger. The electromagnetic, electroweak, neutrino, strong, anomaly, running-continuity, and no-fit provenance branches are checked as ledger objects before they are compared with diagnostic references. The validation package establishes reproducibility of the encoded formulas and their provenance labels. Numerical agreement alone proves

neither derivation nor uniqueness. The neutral C_3 generator is instead justified by the canonical minimal-response theorem established in Theorem 5.4.

Reproducibility statement. The PR-III ledger is supported by step-specific validation scripts, Python builders, and locked JSON artifacts. The harness ledger verifies the inheritance and alpha baseline, the scalar Hessian and tree electroweak block, gauge null removal, and the radiative determinant. It further audits the alpha radiative extraction with D1 and D2 refinements, electroweak scale transport alongside C1 and C2, and the arithmetic of the canonical C3 closure. The proof below, rather than the arithmetic check, establishes the uniqueness of its normalized response functional. The package also checks neutrino branch radiative stability including the $m_{\beta\beta}$ envelope, strong-sector phase-fiber calculations, global running continuity, and the anomaly registry. Target observables are governed by a no-fit provenance manifest and enter only as post-generation diagnostic (DIAG) references. These excluded generation inputs include reference values for the fine-structure constant (α), M_W , M_Z , G_F , neutrino mass-squared splittings (Δm_{21}^2 , Δm_{31}^2), PMNS mixing angles, cosmological neutrino mass bounds, and the strong coupling reference α_3 . The code, Markdown reports, and generated JSON outputs are maintained in the public repository: github.com/oshetskiresearch/Projection_Relativity.

2 Radiative and Cross-Sector Closure in PR-III

We begin by using the PR-II compact electroweak and flavor endpoint, where PR-III extends the projection ledger from tree-level geometric structure to radiative and cross-sector precision. The inherited inputs are not returned. They are used as fixed boundary data for the electromagnetic, electroweak, neutrino, strong, anomaly, and running-continuity branches. The result is a finite no-fit closure chain in which each sector is generated from inherited PR data and earlier PR-generated objects, and only afterward compared with diagnostic references. For a sector s , the construction has the schematic form

$$\mathcal{C}_s : (\mathcal{I}_{\text{PR-I/II}}, \mathcal{G}_{<s}) \longrightarrow \mathcal{G}_s, \quad (4)$$

where $\mathcal{I}_{\text{PR-I/II}}$ denotes the frozen PR-I/PR-II projection ledger and $\mathcal{G}_{<s}$ denotes earlier PR-III generated quantities. Diagnostic references enter only through the later comparison map

$$\mathcal{R}_s = \mathcal{D}_s(\mathcal{G}_s; \mathcal{E}_s), \quad \mathcal{E}_s \cap \text{Dom}(\mathcal{C}_s) = \emptyset. \quad (5)$$

This is the no-fit condition used throughout the paper. Agreement with a reference value is therefore interpreted as a post-generation diagnostic result, not as a parameter fit.

2.1 Summary of PR-III Closure Results

Before detailing the derivations, Table 2 summarizes the major PR-III closures. These milestones should be interpreted as the principal physical achievements of the paper, not merely as a computational log. Although our public repository provides the exact numerical provenance of every result, the underlying physics is governed strictly by the projection sequence developed in the following sections.

Table 2: Principal PR-III closures starting from the PR-II endpoint.

Sector	PR-II endpoint used as input	PR-III extension	Diagnostic status
Electromagnetic	Residual compact electromagnetic boundary $U(1)_{\text{em}}$ and PR boundary constants	Boundary-leakage radiative correction with mode-resolved $D_1 + D_2$ closure	$\alpha_{\text{PR}}^{-1} = 137.035999207513\dots$, $+0.1376\sigma$ against the selected current reference
Electroweak	Compact electroweak lift $\mathbb{R}_w \times S_Y^1 \times SU(2)_L$, weak mixing, and displacement scale	Electromagnetic-to-weak transport, weak-mass reconstruction, vector self-energy, and neutral-channel refinement	$M_W : +0.3566\sigma$, $M_Z : +0.0364\sigma$, $G_F : +0.6003\sigma$
Neutrino	PR-II normal-ordering and PMNS-linked branch	Multiplicative radiative stability envelope, ordering audit, PMNS unitarity rule, and $m_{\beta\beta}$ propagation	Normal ordering preserved; $\sum m_\nu = 0.058733836680 \text{ eV}$; $m_\beta = 8.8150 \text{ meV}$; $m_{\beta\beta} = 1.5077 \text{ meV}$
Strong	Compact color-fiber seed, $N_c = 3$, and PR quark threshold sheets	$SU(3)_c$ identity lock, ordered active-flavor thresholds, and one-loop color-sign audit	$\alpha_{3,\text{PR}} = 0.117861507469150$, diagnostic-compatible at -0.03265% relative residual
Global structure	Completed electromagnetic, electroweak, neutrino, and strong ledgers	Anomaly registry, running-continuity audit, no-fit provenance manifest, and cross-sector diagnostic table	Local anomalies cancel per generation; Witten anomaly absent; residual $U(1)_{\text{em}}$ vectorlike; running continuity and no-fit provenance pass

2.2 Electromagnetic Closure: Precision α result

The first extension is electromagnetic. PR-II identifies the residual electromagnetic phase as the unbroken compact boundary of the electroweak lift. PR-III turns that boundary statement into a radiative precision result. The high-resolution tree baseline is fixed before comparison; the charged-mode inventory is evaluated by the electromagnetic Q^2 trace; neutral modes are excluded from the direct charged trace; and the implemented finite-tier spectral kernel, coefficient, weight, and subtraction convention are fixed as part of the admissible PR-III class before evaluation. The boundary-leakage contribution is

$$D_1 = -p_1^{\text{HR}} \Delta_{\text{EW}}^{\text{HR}} \sin^2 \theta_W^{\text{PR}}. \quad (6)$$

The three-sheet refinement is fixed by the inherited generation count,

$$\epsilon_2^{\text{PR}} = \frac{\Delta_{\text{EW}}^{\text{HR}}}{\sqrt{N_{\text{gen}}^{\text{PR}}}}, \quad D_2 = D_1 \epsilon_2^{\text{PR}}. \quad (7)$$

The generated inverse fine-structure value is therefore

$$\alpha_{\text{PR,phys}}^{-1} = \alpha_{\text{PR,tree}}^{-1} + 4\pi(D_1 + D_2). \quad (8)$$

Numerically, PR-III obtains

$$\begin{aligned}\alpha_{\text{PR,phys}}^{-1} &= 137.035999207513414958891346101616 \\ &5946356776374900155369785562685848 \\ &55576779314655494829901.\end{aligned}\tag{9}$$

This numerical resolution is a direct output of the high-resolution spectral audit and is maintained to ensure exact byte-for-byte reproducibility across the closure chain. Only after this number is generated is it compared with the external fine-structure reference. The residual against the selected current reference is

$$1.5134 \times 10^{-9}, \quad \frac{\Delta\alpha^{-1}}{\sigma} = +0.1376.\tag{10}$$

The compact electromagnetic boundary becomes a current-precision no-fit result, rather than remaining only a tree-level phase normalization.

2.3 Electroweak Closure: Transporting The Electromagnetic Result Into The Weak Scale

The second extension transports this generated electromagnetic precision directly into the weak scale. Building upon the electroweak parent fiber and the residual locking map supplied by PR-II,

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}, \quad Q = T_3 + \frac{Y}{2},\tag{11}$$

PR-III propagates the derived electromagnetic coupling to the weak boundary. This allows for the exact reconstruction of the continuous electroweak couplings,

$$e_{\text{PR}}, \quad g_{\text{PR}} = \frac{e_{\text{PR}}}{\sqrt{\sin^2 \theta_W^{\text{PR}}}}, \quad g'_{\text{PR}} = \frac{e_{\text{PR}}}{\sqrt{1 - \sin^2 \theta_W^{\text{PR}}}},\tag{12}$$

which subsequently generate the weak-mass ledger based solely on the inherited electroweak displacement scale:

$$M_W^2 = \frac{1}{4} g_{\text{PR}}^2 v_{\text{EW,PR}}^2, \quad M_Z^2 = \frac{1}{4} (g_{\text{PR}}^2 + g_{\text{PR}}'^2) v_{\text{EW,PR}}^2, \quad G_F = \frac{1}{\sqrt{2} v_{\text{EW,PR}}^2}.\tag{13}$$

The evaluation proceeds through three constrained refinements: the C1 transport establishes baseline weak-mass consistency, the C2 correction introduces the charged and vector self-energy dynamics, and the C3 branch applies the generation-weighted D_3 term. The charged-adjoint algebra fixes the group factor in D_3 exactly, and the canonical minimal-response theorem in Sec. 5.5 uniquely maps it to the normalized physical- Z correction. Evaluated sequentially on that target-independent closure branch, these yield:

$$M_W^{\text{PR}} = 80.373942851862121269660365960912048165641419593354 \text{ GeV},\tag{14}$$

$$M_Z^{\text{PR}} = 91.187676431043922257304294605837072137923832857910 \text{ GeV},\tag{15}$$

$$G_F^{\text{PR}} = 1.1663792201759949366003257867488874906911567673 \times 10^{-5} \text{ GeV}^{-2}.\tag{16}$$

When evaluated against empirical targets, the corresponding diagnostic distances are

$$M_W : +0.3566\sigma, \quad M_Z : +0.0364\sigma, \quad G_F : +0.6003\sigma.\tag{17}$$

These values establish the stated diagnostic performance of the PR-III electroweak closure branch. The selected diagnostic targets are not inputs to its formulas. The neutral C_3 value is a theorem relative to the declared minimal-response class. A complete Hessian presentation is an alternative microscopic representation of the same response equation and introduces no additional admissible coefficient.

2.4 Neutrino Closure: Radiative Stability Of The Inherited Branch

The third extension tests the PR-II neutrino branch under PR-III radiative stability. The branch is fixed before comparison with oscillation data, cosmological mass ceilings, direct beta-decay bounds, or neutrinoless-double-beta limits. The radiative kernel is used as a stability envelope, not as a fitted deformation. The final generated branch is

$$(m_1, m_2, m_3) = (0, 0.008627283010, 0.05010655367) \text{ eV}, \quad (18)$$

with

$$\sum_i m_i = 0.058733836680 \text{ eV}, \quad m_\beta = 0.008815029409997968 \dots \text{ eV}, \quad m_{\beta\beta} = 0.001507694477 \text{ eV}. \quad (19)$$

Normal ordering remains stable under the multiplicative envelope. The effective Majorana envelope is narrow,

$$1.507519780849 \text{ meV} \leq m_{\beta\beta} \leq 1.507869173151 \text{ meV}, \quad (20)$$

and the PMNS [12, 8] update is constrained to a unitary generator,

$$U_{\text{PMNS}}^{\text{phys}} = U_{\text{PMNS}}^0 \exp(A_\nu^{\text{PR}}), \quad (A_\nu^{\text{PR}})^\dagger = -A_\nu^{\text{PR}}. \quad (21)$$

The neutrino branch therefore remains a stable normal-ordering projection branch whose mass sum, beta observable, and $m_{\beta\beta}$ scale remain below the current diagnostic ceilings.

2.5 Strong-Sector Closure: Compact Color Fiber and Active Flavor

The fourth extension turns to the strong sector, firmly anchoring the compact color-fiber branch. PR-III establishes the geometric constraints on the $SU(3)_c$ gauge group, securing the standard algebraic constants:

$$N_c = 3, \quad C_A = 3, \quad C_F = \frac{4}{3}, \quad T_F = \frac{1}{2}, \quad (22)$$

Simultaneously, we preserve the strictly ordered hierarchy of active-flavor thresholds derived from the internal projection sheets:

$$u < d < s < c < b < t. \quad (23)$$

Evaluated at the generated electroweak matching scale of $\Lambda_{\text{EW}}^{\text{PR}} = 173.67059834472832 \text{ GeV}$, the framework naturally saturates the active-flavor count at

$$n_f^{\text{PR}} = 6. \quad (24)$$

This comprehensively determines the reduced one-loop color coefficient,

$$\beta_0^{\text{PR}} = 11 - \frac{2}{3}n_f^{\text{PR}} = 7, \quad (25)$$

ensuring that the required negative sign for asymptotic freedom is securely preserved across all allowed active-flavor intervals. When evaluated dynamically, the projection establishes a final strong coupling anchor of

$$\alpha_{3,\text{PR}} = 0.117861507469150. \quad (26)$$

Comparing this theoretically derived baseline to the external diagnostic target yields a relative residual of:

$$\alpha_{3,\text{PR}} - \alpha_s^{\text{diag}} = -0.000038492530850, \quad \frac{\Delta\alpha_3}{\alpha_s^{\text{diag}}} = -0.0326484570399\%. \quad (27)$$

This calculation is the closed theorem for the declared compact color-fiber, sharp-threshold, one-loop domain. It verifies the active-threshold ordering and one-loop asymptotic-freedom sign in that class. All-orders or nonperturbative QCD structures define a different domain and are not alternative solutions of the displayed PR-III equation. The displayed coupling comparison is post-generation only and neither fixes the strong anchor nor defines an admissibility tier.

2.6 Cross-Sector Audits: Anomaly, Continuity, and Provenance

The final extension compares the declared finite-tier branches through three cross-sector structural audits:

- **Anomaly Registry:** Verifies that local gauge anomalies cancel per generation, the Witten $SU(2)$ anomaly is absent, and the residual electromagnetic branch remains purely vector-like.
- **Running Continuity:** Confirms that the electromagnetic-to-electroweak, electroweak-to-neutrino, electroweak-to-strong, and anomaly-continuity links seamlessly integrate without breaking the generated ledger.
- **No-fit Provenance Manifest:** Ensures that forbidden target observables remain strictly excluded from generation, entering only as post-generation diagnostic references.

With these audits satisfied, the following finite-tier statuses are recorded:

- **Precision Status:** The fundamental observables (α^{-1} , M_W , M_Z , and G_F) successfully pass one-sigma diagnostic thresholds.
- **Compatibility Status:** The neutrino and strong branches remain radiatively stable and highly compatible with current diagnostic ceilings.
- **Structural Status:** Anomaly cancellation, running continuity, and strict no-fit provenance are fully preserved.

PR-III therefore provides a reproducible, no-fit, cross-sector ledger at the declared finite tier. Its theorem domain is exactly the stated PR radiative and cross-sector construction. Higher radiative orders, different threshold profiles, additional fields, and independent EFT operators change that domain and are therefore outside the PR-III solution class, not unresolved solutions within it. The completed equation quotient is stated in Sec. 9.

2.7 Organization of the Derivations

The remainder of this manuscript evaluates the finite-tier relations and exact structural identities summarized above. We begin by freezing the inherited PR-I and PR-II geometric baselines used for the PR-III fine-structure calculation. Following this, the framework examines the scalar displacement Hessian and gauge-quotient structure used to isolate the electroweak null directions.

With the inherited baseline frozen, the analysis proceeds sequentially through the phenomenological sectors. The electromagnetic calculation evaluates the declared boundary-leakage and mode-resolved $D_1 + D_2$ relation. This finite-tier result is transported to the weak scale through the declared weak-mass and neutral-channel ledgers. The neutrino section proves stability under its stated multiplicative envelope, while the strong-sector section verifies the compact color identities, threshold ordering, and one-loop sign. PR-IV then applies the common admissibility and equivalence rules to the completed selectors and proves that their equation quotient is a singleton.

The calculations culminate in a cross-sector audit of anomaly cancellation, running continuity, and strict no-fit provenance across the implemented branches. The manuscript concludes by recording the exact closed theorem domain and the exclusion of every inequivalent in-class equation.

3 Inherited Baseline and Alpha Policy

We now define the inherited baseline used by the alpha calculation and the electroweak seed refined in later sections. Its role is to fix the geometry, the high-resolution tree value, and the no-fit rule before any precision comparison is made.

3.1 Inherited Geometric and Projection Baselines

The theoretical foundation of this framework rests upon the universal PR master field,

$$\Psi(x, \xi), \quad (28)$$

defined over observable spacetime and the internal projection manifold. The internal geometry inherited from PR-I is given by

$$\mathcal{M}_{\text{int}} = \mathbb{R}_w \times S_\theta^1, \quad (29)$$

which admits the separable internal spectral operator

$$O_{\text{int}} = O_X + O_\theta. \quad (30)$$

The radial branch, strictly preserved from PR-I, takes the form

$$O_X = -\frac{d^2}{dw^2} + 1 + w^2 + \frac{3}{4}w^4, \quad (31)$$

wherein the quadratic coefficient is strictly determined by the unit local radial stiffness, and the quartic coupling is dictated by the projection trace. Within the locked, finite-resolution PR-I framework, this configuration yields a fixed radial spectral gap of

$$\mu_{\text{min}}^2 = 3.052966743096. \quad (32)$$

Correspondingly, the compact phase dynamics are governed by the operator

$$O_\theta = -\frac{1}{R_A^2} \frac{d^2}{d\theta^2}, \quad (33)$$

which furnishes the residual electromagnetic phase normalization. Sourced directly from the released finite-resolution preprint ledger, the inherited compact-boundary stiffness and its corresponding cofactor are fixed at

$$q_{\text{bc}} = 10.905007182855176, \quad c_{\text{bc}} = 0.796684464847899, \quad (34)$$

yielding a boundary-resolved electromagnetic normalization of

$$(\alpha_{\text{PR,bc}})^{-1} = 137.036361812007. \quad (35)$$

We present this value as the tree-level phase baseline; it does not represent the final, radiatively corrected PR-III precision value for the fine-structure constant. From PR-II, the framework incorporates the compact electroweak lift, defined over the extended manifold

$$\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L, \quad (36)$$

associated with the standard residual charge operator

$$Q = T_3 + \frac{Y}{2}. \quad (37)$$

The fundamental electroweak boundary data initialized for PR-III comprise the geometric displacement and the weak mixing angle,

$$\Delta_{\text{EW}} = 0.018644236701811, \quad \sin^2 \theta_W^{\text{PR}} = 0.231355763298189, \quad (38)$$

alongside the tree-level custodial parameter

$$\rho_{\text{EW}}^{(0)} = 1. \quad (39)$$

Finally, the internal generation count is inherited as

$$N_{\text{gen}}^{\text{PR}} = 3. \quad (40)$$

This topological integer plays a critical role in the subsequent PR-III derivations, acting as a primary geometric constraint for both the mode-resolved fine-structure refinement and the neutral electroweak corrections.

3.2 High-Resolution Asymptotic Tree Baseline

Transitioning from the structural classification of the PR-I and PR-II ledgers to the precision phenomenology examined in PR-III requires a formally strict geometric background. The previously released preprint value,

$$(\alpha_{\text{PR,bc}})^{-1} = 137.036361812007, \quad (41)$$

was sufficient for finite-resolution topological classification. However, evaluating quantum-scale radiative corrections fundamentally requires an unperturbed tree baseline whose inherent numerical resolution strictly exceeds the magnitude of the anticipated perturbative shift. To assign physical meaning to the radiative sector, PR-III must therefore evaluate the alpha branch using the exact high-resolution asymptotic convergence limit:

$$(\alpha_{\text{PR,tree}})^{-1} = 137.036041314016444. \quad (42)$$

This baseline is locked prior to any radiative calculation. It is derived exclusively from the geometric convergence sequence of the PR architecture, entirely independent of external diagnostic fitting. This same high-resolution baseline concurrently locks the corresponding geometric boundary parameters:

$$q_{\text{bc}}^{\text{HR}} = 10.904981678435453, \quad c_{\text{bc}} = 0.796684464847899, \quad p_1^{\text{HR}} = 7.685346950896231 \times 10^{-4}, \quad (43)$$

along with the refined radial spectral gap,

$$\left(\mu_{\text{min}}^{\text{HR}}\right)^2 = 3.053016666732689. \quad (44)$$

The shift from the previous finite-resolution boundary value to this high-resolution tree limit is mathematically exact:

$$(\alpha_{\text{PR,tree}})^{-1} - (\alpha_{\text{PR,bc}})^{-1} = -3.20497990556 \times 10^{-4}. \quad (45)$$

It is imperative to emphasize that this displacement is not a radiative physical correction; it is purely the geometric convergence refinement of the tree-level boundary ledger. Only after Eq. (42) is formally fixed does the architecture permit the evaluation of radiative physical corrections.

3.3 Strict Predictive Constraints and Diagnostic Firewall

To ensure the integrity of the theoretical derivation, PR-III enforces a strict predictive firewall against empirical tuning. Let α_{ref}^{-1} denote an external empirical fine-structure reference used for terminal diagnostic comparison. The PR-III derivation imposes

$$\alpha_{\text{ref}}^{-1} \notin \text{Dom}(\mathcal{C}_\alpha), \quad (46)$$

where $\mathcal{C}_\alpha$ is the formally defined PR fine-structure construction map. The mathematically permissible inputs for generation are restricted exclusively to inherited structural data. These inputs consist solely of the boundary quantities, the high-resolution tree baseline, the weak boundary deficit, the generation count, the charged-mode inventory, and the PR spectral kernel. The empirical diagnostic target is absolutely forbidden from entering the generation domain. Under this constrained topology, the physical alpha branch takes the form

$$(\alpha_{\text{PR,phys}})^{-1} = (\alpha_{\text{PR,tree}})^{-1} + \Delta\alpha_{\text{PR,rad}}^{-1}, \quad (47)$$

where the radiative shift $\Delta\alpha_{\text{PR,rad}}^{-1}$ must emerge dynamically from native PR geometry. The diagnostic reference is permitted to enter the analysis only after Eq. (47) has been fully evaluated.

To establish the required scale of this physical correction, we note that the selected empirical diagnostic reference utilized in the final precision audit is

$$\alpha_{\text{diag}}^{-1} = 137.035999206. \quad (48)$$

While this value is strictly excluded from defining $\Delta\alpha_{\text{PR,rad}}^{-1}$, it provides the phenomenological target. Relative to the high-resolution tree baseline, the empirical displacement is

$$\alpha_{\text{diag}}^{-1} - (\alpha_{\text{PR,tree}})^{-1} = -4.2108016444 \times 10^{-5}. \quad (49)$$

The central objective of the subsequent electromagnetic derivations is to demonstrate that native radiative boundary leakage naturally generates a physical correction of precisely this scale, emerging entirely from first principles without phenomenological adjustment.

3.4 Tree-Level Electroweak Seed and Pre-Radiative Initialization

The high-resolution electromagnetic baseline established above acts as the geometric seed for the electroweak sector, fixing the exact initial state prior to weak-scale transport and neutral-channel refinement. Propagating the exact tree-level fine-structure value,

$$\alpha_{\text{PR,tree}} = (137.036041314016444)^{-1}, \quad (50)$$

the architecture analytically reconstructs the bare, pre-radiative gauge couplings:

$$e_{\text{PR}} = 0.30282207421182286581809072020845327913143244571434, \quad (51)$$

$$g_{\text{PR}} = 0.62957484689764151178346495625282724825379638851140, \quad (52)$$

$$g'_{\text{PR}} = 0.34540199548414150920957567770984563383700364736773. \quad (53)$$

Coupling this continuous gauge sector to the inherited electroweak displacement scale,

$$v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV}, \quad (54)$$

the framework generates the corresponding tree-level weak-mass baseline:

$$M_{W,0}^{\text{PR}} = 77.506832192893635832893671080594197560828951119835 \text{ GeV}, \quad (55)$$

$$M_{Z,0}^{\text{PR}} = 88.40509587274585732715680415742085114480023483002 \text{ GeV}, \quad (56)$$

$$\rho_{\text{EW}}^{(0)} = 1. \quad (57)$$

We note that these initial mass parameters do not represent the final phenomenological predictions of the theory. Rather, they constitute the mathematically exact pre-radiative geometric seed. As demonstrated in the ensuing derivations, the C1, C2, and C3 electroweak branches dynamically transport and refine this seed, incorporating necessary self-energy dynamics and neutral-channel corrections to elevate the baseline into the final, high-precision electroweak physical observables.

3.5 Gauge-Null Quotient and Physical Scalar Isolation

To evaluate and interpret the radiative determinant, it is mathematically necessary to isolate the physical scalar degrees of freedom from the unphysical gauge-null directions. Expanding the scalar sector in the standard basis,

$$(h_A, \pi_1, \pi_2, \pi_3), \quad (58)$$

the corresponding symbolic Hessian spectrum evaluates to

$$\{H_{AA}, 0, 0, 0\}. \quad (59)$$

The three degenerate null directions correspond exactly to the Goldstone modes associated with the broken gauge quotient,

$$(SU(2)_L \times U(1)_Y)/U(1)_{\text{em}}, \quad (60)$$

leaving the physical scalar quotient entirely restricted to the radial amplitude direction, h_A . The dynamical stability of this symmetry-breaking configuration is strictly guaranteed by the positive-definite curvature condition,

$$H_{AA} = 8\beta_A A_0^2 > 0, \quad (61)$$

which requires both $\beta_A > 0$ and $A_0^2 > 0$.

The neutral gauge sector geometry is locked to enforce a massless photon trajectory orthogonal to a massive, positive-definite Z direction. Projected into the $[W^3, B]$ gauge basis, the inherited weak mixing angle provides the rotational coefficients:

$$\sin \theta_W^{\text{PR}} = 0.480994556412220740217768765489223953539786055779711286076093, \quad (62)$$

$$\cos \theta_W^{\text{PR}} = 0.876723580555360105457398073670122851235684044309931260035474. \quad (63)$$

By enforcing that the photon eigenvalue remains zero while the tree-level Z eigenvalue is only positive, this quotient procedure formally prevents the subsequent radiative integration from artificially conflating unphysical gauge-null artifacts with genuine physical scalar curvature.

With the physical scalar and gauge-null directions separated, the inherited baseline used by the finite-tier fine-structure calculation is now frozen for reproducibility. This initialization fixes the PR-I spectral operators and boundary parameters, the declared $N_{\text{gen}}^{\text{PR}} = 3$ realization, the pre-radiative electroweak seed, and the high-resolution tree baseline of $137.036041314016444 \dots$. Empirical references are excluded from the generation map. This makes the implemented determinant, charged-mode inventory, spectral kernel, weights, and subtractions a fixed no-fit finite-tier class. The resulting $D_1 + D_2$ value is compared with data only as a post-generation diagnostic; changing any registered kernel or subtraction defines a different class rather than a second solution of this one.

4 Alpha and Electromagnetic Radiative Evaluation

The preceding section fixed the inherited PR-I/PR-II boundary data, the high-resolution tree value of the compact electromagnetic normalization, and the exclusion of the diagnostic fine-structure reference from the generation map. This section evaluates the declared PR-III electromagnetic relation. It does not fit α , and the later diagnostic displacement is not used to select the equation, coefficient, or operator class.

The starting point is the high-resolution tree baseline

$$(\alpha_{\text{PR,tree}})^{-1} = 137.036041314016444, \quad (64)$$

with the physical branch written as

$$(\alpha_{\text{PR,phys}})^{-1} = (\alpha_{\text{PR,tree}})^{-1} + \Delta\alpha_{\text{PR,rad}}^{-1}. \quad (65)$$

The radiative term is generated from the compact electroweak boundary deficit, the charged-mode inventory, and the PR spectral kernel. The diagnostic fine-structure reference enters only after this construction has been evaluated.

4.1 Radiative Determinant and Extraction Convention

The one-loop PR radiative architecture is organized by a reference-normalized supertrace determinant,

$$\Delta\Gamma_{\text{PR}}^{(1)}[B] = \frac{1}{2} \text{STr}'_{\text{PR}} \log [H_{\text{PR}}[B] H_{\text{PR}}[0]^{-1}]. \quad (66)$$

The prime indicates that gauge-null modes and exact zero modes are either quotient-removed or reference-normalized. In block form,

$$\begin{aligned} \Delta\Gamma_{\text{PR}}^{(1)} = & \frac{1}{2} \text{Tr}'_{\text{scalar}} \log(H_A/H_{A0}) + \frac{1}{2} \text{Tr}'_{\text{vector}} \log(O_V/O_{V0}) \\ & - \text{Tr}'_{\text{ghost}} \log(M_{\text{gh}}/M_{\text{gh}0}) - \text{Tr}'_{\text{fermion}} \log(D_F/D_{F0}). \end{aligned} \quad (67)$$

The physical scalar block is the quotient-retained radial amplitude direction $H_A = H_{AA}$, while the gauge-null scalar directions have already been removed. The photon is treated as the residual massless $U(1)_{\text{em}}$ background, not as a massive internal determinant mode. The charged weak-vector, ghost, charged Goldstone partner, lepton, and quark sectors enter only through their allowed charged contributions.

The electromagnetic extraction convention is fixed by the coefficient of the residual background kinetic term,

$$\Delta\Gamma_{\text{PR}}^{(1)}[F] \supset -\frac{1}{4} \Delta Z_A^{\text{PR}} \int d^4x F_{\mu\nu} F^{\mu\nu}. \quad (68)$$

The corresponding shift in the inverse fine-structure normalization is

$$\Delta\alpha_{\text{PR,rad}}^{-1} = 4\pi \Delta Z_A^{\text{PR}}. \quad (69)$$

This convention is fixed before the numerical equation is evaluated.

4.2 Charged-mode Inventory

Only modes carrying residual electromagnetic charge contribute directly to ΔZ_A^{PR} . The generic charged-mode structure is

$$\Delta Z_A^{\text{PR}} = \sum_i \sigma_i n_i Q_i^2 K_i^{\text{PR}}, \quad (70)$$

where σ_i is the supertrace sign, n_i is the multiplicity, Q_i is the residual electromagnetic charge, and K_i^{PR} is the PR spectral kernel for that sector.

The charged weak vectors $W^\pm$ contribute a charge-square sum of 2. The charged electroweak ghost pair also carries charge-square sum 2, with the opposite ghost sign required by the gauge convention. Gauge-fixing Goldstone partners may appear only as conditional partners of the vector/ghost block and are not counted as independent physical scalar modes. The charged leptons contribute charge-square sum 3. The color-resolved up-type quarks contribute 4, and the color-resolved down-type quarks contribute 1. Thus the charged fermion charge-square sum is

$$\sum_{\text{charged fermions}} Q_i^2 = 8, \quad (71)$$

or 8/3 per generation. Neutral modes, including the photon background, Z , neutrinos, the neutral radial scalar, and gluons, have no direct Q^2 contribution to the present electromagnetic extraction.

4.3 PR Spectral Kernel

To evaluate the physical radiative shift generated by the charged-mode inventory, the framework integrates over the internal fluctuation spectrum. The declared finite-tier construction represents that integration by a fixed PR spectral-kernel class. Its heat-kernel coefficients, weights, subtraction functions, domain, and power counting are registered as part of the admissibility definition. The kernel is

$$K_i^{\text{PR}} = \frac{C_i^{\text{hk}}}{16\pi^2} I_i^{\text{PR}}, \quad \text{with} \quad \Delta Z_A^{\text{PR}} = \sum_i \sigma_i n_i Q_i^2 K_i^{\text{PR}}. \quad (72)$$

The regulator is not an arbitrary cutoff. It is tied to the internal PR spectral gap. With

$$\mu_n^2 = \lambda_n - \lambda_0, \quad \mu_n^2 \geq \left(\mu_{\min}^{\text{HR}}\right)^2 > 0, \quad (73)$$

the admissible kernel class has the schematic form

$$I_i^{\text{PR}}(\tau_i) = \sum_{n \in S_i} \omega_{i,n}^{\text{PR}} \left[\log \left(1 + \frac{\tau_i}{\mu_n^2} \right) - R_i(\tau_i; \mu_n^2) \right]. \quad (74)$$

For the electromagnetic closure branch the reference scale is not a free normalization. It is fixed to the inherited weak displacement anchor,

$$\Lambda_{\text{ref}} = \Lambda_{\text{EW}}^{\text{PR}} = 173.67059834472832 \text{ GeV}. \quad (75)$$

The threshold variable is therefore

$$\tau_i = \frac{M_i^2}{(\Lambda_{\text{EW}}^{\text{PR}})^2}. \quad (76)$$

The allowed masses M_i are likewise restricted to frozen PR values, PR-II generated mass-sector outputs, PR-III derived masses, or symbolic placeholders awaiting PR derivation. Empirical masses, fitted thresholds, or values chosen to force the fine-structure residual are forbidden as generation inputs. For the alpha branch the locked boundary anchors are

$$p_1^{\text{HR}} = 7.685346950896231 \times 10^{-4}, \quad q_{\text{bc}}^{\text{HR}} = 10.904981678435453, \quad c_{\text{bc}} = 0.796684464847899. \quad (77)$$

The high-resolution weak boundary deficit is therefore

$$\Delta_{\text{EW}}^{\text{HR}} = \frac{1 - c_{\text{bc}}}{q_{\text{bc}}^{\text{HR}}} = 0.018644280306692899718928776428403028768291215822597. \quad (78)$$

4.4 Minimal Boundary-Leakage Correction

The calculation of the radiative fine-structure shift starts by isolating the primary geometric contribution, formally designated as the minimal boundary-leakage term. This correction arises from the interplay among the inherited structural parameters of the high-resolution radial boundary anchor, the refined weak boundary deficit, and the native weak mixing angle. The framework formally defines this leading-order leakage as:

$$D_1 = -p_1^{\text{HR}} \Delta_{\text{EW}}^{\text{HR}} \sin^2 \theta_W^{\text{PR}}. \quad (79)$$

Using the frozen weak-angle output and Eq. (78), PR-III obtains

$$D_1 = -3.3150449735433222932326442973740154803469215638087 \times 10^{-6}. \quad (80)$$

The induced inverse-alpha correction is

$$\Delta \alpha_{\text{PR},D1}^{-1} = 4\pi D_1 = -4.1658083740813887216904843821118454991063065733008 \times 10^{-5}, \quad (81)$$

so the minimal physical branch is

$$(\alpha_{\text{PR},D1})^{-1} = 137.035999655932703186112783095156178881545008936934266992. \quad (82)$$

Compared only afterward with the selected diagnostic reference, this first correction accounts numerically for 98.9314797011% of the difference between the frozen tree value and that reference and leaves the diagnostic residual

$$4.49932703186112783095156178881545008936934266992 \times 10^{-7}. \quad (83)$$

This comparison is reserved exclusively for post-generation diagnostic assessment. The structural sign, scale, and admissible class are defined by the PR construction independently of that comparison.

4.5 Three-Sheet Mode Refinement

The finite-tier electromagnetic branch next applies the target-independent three-sheet scaling built from the inherited generation count, $N_{\text{gen}}^{\text{PR}} = 3$. Within the declared three-sheet closure class, the square-root multiplicity rule is

$$\epsilon_2^{\text{PR}} = \frac{\Delta_{\text{EW}}^{\text{HR}}}{\sqrt{N_{\text{gen}}^{\text{PR}}}}, \quad N_{\text{gen}}^{\text{PR}} = 3. \quad (84)$$

Numerically, we find the values to be:

Evaluating the declared mode-resolved boundary-leakage corrections on the frozen high-resolution tree baseline gives the following finite-tier result. Its inputs are the inherited PR internal geometry, structural boundary deficit, and declared three-sheet scaling, and none is selected from the diagnostic fine-structure value. This establishes reproducibility and no-fit provenance for the displayed calculation. Under the completed selector rules proved in PR-IV, alterations of the registered D_1/D_2 operator data are out of class, while equivalent rewritings represent the same equation:

$$(\alpha_{\text{PR,phys}})^{-1} = 137.03599920751341495889134610161659463567763749$$

$$0015536978556268584855576779314655494829901.$$

(85)

Table 3: Mode-resolved boundary leakage corrections and inverse fine-structure evaluation.

Radiative Term	Generated Value
ϵ_2^{PR}	1.0764280253915984059993061466845953623 291973168297815611743348884979166980206478 $\times 10^{-2}$
$D_2 = D_1 \epsilon_2^{\text{PR}}$	-3.5684073149555819954984631935783837502999439569 8611894343571143823853634129916305743510899 $\times 10^{-8}$
$\Delta Z_{A,D_1 D_2}^{\text{PR}} = D_1 + D_2$	-3.350729046692878113187628929309799317849 92100337851630786204253710293778758370074941617298 $\times 10^{-6}$
$\Delta \alpha_{D_1 D_2}^{-1} = 4\pi(D_1 + D_2)$	-4.2106503029041108653898383405364322362509 9844630214437314151444232206853445051700990393453 $\times 10^{-5}$

4.6 Precision Status

To evaluate the predictive fidelity and numerical accuracy of the derived fine-structure constant, we subject the first-principles geometric output to an empirical confrontation. For this baseline comparison, the selected current empirical diagnostic reference and its associated experimental uncertainty are given by:

$$\alpha_{\text{diag}}^{-1} = 137.035999206, \quad \sigma_{\text{diag}} = 1.1 \times 10^{-8}. \quad (86)$$

This reference has not entered the construction of Eq. (85). It is used only for the post-generation diagnostic residual,

$$\Delta_{\text{diag}} = (\alpha_{\text{PR,phys}})^{-1} - \alpha_{\text{diag}}^{-1} = 1.51341495889134610161659463567763749001553 \\ 6978556268584855576779314655494829901 \times 10^{-9}. \quad (87)$$

Equivalently,

$$\boxed{\frac{\Delta_{\text{diag}}}{\sigma_{\text{diag}}} = +0.1376,} \quad (88)$$

so the generated value lies inside the selected one-sigma diagnostic band. As a post-generation descriptive comparison only, the $D_1 + D_2$ branch accounts for

$$\boxed{99.9964058744945\%} \quad (89)$$

of the high-resolution tree-to-diagnostic displacement. Relative to the CODATA 2022 [13] recommended-value comparison, the same generated branch remains inside two sigma.

4.7 Finite-Tier Status of the Electromagnetic Branch

The PR-III electromagnetic equation is reproducibly evaluated at the declared finite tier. The residual compact electromagnetic boundary inherited from PR-I/PR-II is carried from the tree normalization to the following no-fit finite-tier value:

$$\boxed{(\alpha_{\text{PR,tree}})^{-1} \longrightarrow (\alpha_{\text{PR,phys}})^{-1} = 137.035999207 \dots} \quad (90)$$

The construction uses the high-resolution boundary constants, weak boundary deficit, weak-angle output, generation count, charged-mode supertrace inventory, and internal PR spectral kernel. It does not use the diagnostic fine-structure reference as a generation input.

This closes the electromagnetic equation at the declared tier. The residual against the selected reference is nonzero and lies below the supplied experimental uncertainty; it is a diagnostic output, not an adjustable coefficient. The invariant operator basis, kernel weights, subtractions, and power counting are fixed elements of the PR-III admissibility domain. An all-orders series or a differently regulated kernel is a different theorem domain and is not a competing solution of the finite-tier equation proved here.

5 Electroweak Scale Transport and Precision Closure

The electromagnetic closure obtained in the preceding section supplies the low-energy boundary condition for the weak sector,

$$(\alpha_{\text{PR,phys}}(0))^{-1} = 137.035999207 \dots \quad (91)$$

PR-III now transports this generated quantity to the inherited electroweak displacement scale

$$\Lambda_{\text{EW}}^{\text{PR}} = 173.67059834472832 \text{ GeV}, \quad v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV}. \quad (92)$$

Neither scale is selected from the weak-boson diagnostic values. Both are inherited PR quantities, fixed before the electroweak construction begins. The electroweak calculation proceeds in three branches. The C_1 branch performs the minimal compact-boundary transport and reconstructs the weak couplings and tree-form mass ledger at the electroweak scale. The C_2 branch introduces a generation-weighted charged neutral vector self-energy split. The C_3 branch supplies the next neutral-channel correction while leaving the charged branch unchanged. The diagnostic values of M_W , M_Z , and G_F enter only after each branch has been generated.

5.1 Electromagnetic-to-electroweak Transport

The PR transport equation is written as

$$(\alpha_{\text{PR}}(\mu))^{-1} = (\alpha_{\text{PR}}(0))^{-1} - \Delta_{\text{run}}^{\text{PR}}(0 \rightarrow \mu). \quad (93)$$

At the level of the full spectral construction,

$$\Delta_{\text{run}}^{\text{PR}} = \sum_i B_i^{\text{PR}} L_i^{\text{PR}}, \quad (94)$$

where the inventory preserves the charged-mode content of the electromagnetic supertrace. Charged weak vectors, their gauge partners, charged leptons, and color-resolved quarks contribute; neutral modes have no direct electromagnetic Q^2 contribution. The spectral transport kernel is reference-normalized and tied to the PR gap,

$$\mu_n^2 \geq (\mu_{\text{min}}^{\text{HR}})^2 = 3.053016666732689, \quad (95)$$

with the matching scale fixed to $\Lambda_{\text{EW}}^{\text{PR}}$. Empirical weak masses and fitted thresholds are not admissible transport inputs. The minimal compact-boundary transport relation is

$$\Delta_{\text{run}}^{C_1} = q_{\text{bc}}^{\text{HR}} c_{\text{bc}}, \quad (96)$$

where

$$q_{\text{bc}}^{\text{HR}} = 10.904981678435453, \quad c_{\text{bc}} = 0.796684464847899. \quad (97)$$

This result in

$$\Delta_{\text{run}}^{C_1} = 8.687829492660492292065934163247, \quad (98)$$

and Eq. (93) gives

$$\begin{aligned} \left(\alpha_{\text{PR}}^{C_1} (\Lambda_{\text{EW}}^{\text{PR}}) \right)^{-1} &= 128.348169714852922666825411938369594635 \\ &67763749001553697855626858485557677931466. \end{aligned} \quad (99)$$

The transported electromagnetic coupling and the electroweak couplings are then reconstructed from the inherited weak angle,

$$e_{\text{PR}}^{C_1} = \sqrt{4\pi\alpha_{\text{PR}}^{C_1}} = 0.3129032630130971839408029965751121711\dots, \quad (100)$$

$$g_{\text{PR}}^{C_1} = \frac{e_{\text{PR}}^{C_1}}{\sqrt{\sin^2 \theta_W^{\text{PR}}}} = 0.6505338965726955536662186423635601421\dots, \quad (101)$$

$$g_{\text{PR}}^{\prime C_1} = \frac{e_{\text{PR}}^{C_1}}{\sqrt{1 - \sin^2 \theta_W^{\text{PR}}}} = 0.3569007038853554861535860269366292845\dots, \quad (102)$$

with

$$\sin^2 \theta_W^{\text{PR}} = 0.231355763298189. \quad (103)$$

5.2 The C_1 Weak-Mass Ledger

Having dynamically transported the high-precision electromagnetic fine-structure closure up to the electroweak boundary, the architecture now recouple this refined gauge sector to the invariant geometric displacement scale, $v_{\text{EW}}^{\text{PR}}$. This interaction reconstructs the physical gauge-boson masses and the Fermi coupling from the transported continuous couplings and the native symmetry-breaking kinematics. We designate this initial structural reconstruction as the C_1 baseline weak-mass ledger. Prior to integrating the necessary self-energy and neutral-channel radiative loops, the C_1 relations are formally defined by:

$$\left(M_W^{C_1} \right)^2 = \frac{1}{4} \left(g_{\text{PR}}^{C_1} \right)^2 \left(v_{\text{EW}}^{\text{PR}} \right)^2, \quad (104)$$

$$\left(M_Z^{C_1} \right)^2 = \frac{1}{4} \left[\left(g_{\text{PR}}^{C_1} \right)^2 + \left(g_{\text{PR}}^{\prime C_1} \right)^2 \right] \left(v_{\text{EW}}^{\text{PR}} \right)^2, \quad (105)$$

$$G_F^{\text{PR}} = \frac{1}{\sqrt{2} \left(v_{\text{EW}}^{\text{PR}} \right)^2}, \quad (106)$$

$$\rho_{\text{EW}}^{C_1} = \frac{\left(M_W^{C_1} \right)^2}{\left(M_Z^{C_1} \right)^2 \cos^2 \theta_W^{\text{PR}}}. \quad (107)$$

Numerically,

$$M_W^{C_1} = 80.0870965635111 \text{ GeV}, \quad (108)$$

$$M_Z^{C_1} = 91.3481721488317 \text{ GeV}, \quad (109)$$

$$G_F^{\text{PR}} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}, \quad (110)$$

$$\rho_{\text{EW}}^{C_1} = 1. \quad (111)$$

The C_1 branch is internally consistent and preserves the tree-form ρ identity. Its post-generation comparison is recorded only as a diagnostic: G_F lies within one standard deviation, whereas the W and Z values do not close uniformly. This comparison neither generates nor fixes the next equation. Any charged-neutral refinement must be justified independently by the non-Abelian representation and a target-independent projector-trace or operator-basis theorem.

5.3 Charged and Neutral Vector Branches

Before introducing a self-energy correction, PR-III tests whether a single shift in the weak angle could close both vector masses. Holding the generated electromagnetic coupling and displacement scale fixed, the W diagnostic would require

$$\sin^2 \theta_W|_W = 0.2297344528823705608 \dots, \quad (112)$$

whereas the Z diagnostic would require

$$\sin^2 \theta_W|_Z = 0.2325249689814497325 \dots \quad (113)$$

The required shifts have opposite signs,

$$\Delta \sin^2 \theta_W|_W = -0.0016213104158184392 \dots, \quad (114)$$

$$\Delta \sin^2 \theta_W|_Z = +0.0011692056832607325 \dots, \quad (115)$$

with spread

$$2.79051609907917 \times 10^{-3}. \quad (116)$$

A single weak-angle displacement gives two different post-generation central-value reconstructions. This is a diagnostic distinction within the angle-only comparison class and is not used to infer the form, coefficient, or admissible class of the subsequent C_2 equation. The charged–neutral structure and normalization are defined independently at the representation and canonical-closure level.

5.4 Generation-weighted vector self-energy: the C_2 branch

The declared finite-tier C_2 relation uses the same weak-boundary deficit and generation count that enter the electromagnetic refinement. Its charged–neutral coefficient pair $(2, -1)$ is a target-independent PR closure at the stated tier. Operators with an independent coefficient or a different kernel are excluded by the completed admissibility rules and belong to a different equation class. Define

$$\epsilon_V^{\text{PR}} = \frac{\Delta_{\text{EW}}^{\text{HR}}}{\sqrt{N_{\text{gen}}^{\text{PR}}}}, \quad N_{\text{gen}}^{\text{PR}} = 3, \quad (117)$$

so that

$$\epsilon_V^{\text{PR}} = 0.0107642802539159840599930614668459536 \dots \quad (118)$$

The C_2 vector self-energies are

$$\Pi_W^{C_2} = \frac{2}{N_{\text{gen}}^{\text{PR}}} \epsilon_V^{\text{PR}}, \quad \Pi_Z^{C_2} = -\frac{1}{N_{\text{gen}}^{\text{PR}}} \epsilon_V^{\text{PR}}. \quad (119)$$

Numerically,

$$\Pi_W^{C_2} = 0.0071761868359439894 \dots, \quad \Pi_Z^{C_2} = -0.0035880934179719947 \dots \quad (120)$$

The corrected vector masses are

$$M_W^{C_2} = M_W^{C_1} \sqrt{1 + \Pi_W^{C_2}}, \quad M_Z^{C_2} = M_Z^{C_1} \sqrt{1 + \Pi_Z^{C_2}}, \quad (121)$$

which give

$$M_W^{C_2} = 80.3739428518621 \text{ GeV}, \quad (122)$$

$$M_Z^{C_2} = 91.1841419901086 \text{ GeV}, \quad (123)$$

$$\rho_{\text{EW}}^{C_2} = 1.01080304257989. \quad (124)$$

The C_2 split reduces the absolute W and Z residuals relative to C_1 by factors of approximately 59.48 and 46.43, respectively. At this stage M_W and G_F lie inside one standard deviation, while M_Z lies inside two standard deviations but remains outside one.

5.5 Neutral-channel D_3 : exact group factor and canonical-closure uniqueness

At C_3 the charged branch and photon-null direction are held fixed, so the allowed neutral correction is classified before any diagnostic comparison. The C_2 residual is reported only after this construction and does not select its coefficient. Define

$$\rho_{\text{PR}} := \frac{\left(\epsilon_V^{\text{PR}}\right)^2}{N_{\text{gen}}^{\text{PR}}} > 0 \quad (125)$$

At the declared fixed Lorentz/momentum kernel, equip the real gauge-index coefficient space with the canonical kinetic inner product and take the ordered field basis (W^1, W^2, Z, A) to be orthonormal. Let P_Z denote the resulting orthogonal projector onto the normalized physical Z direction.

Proposition 5.1 (Rank-one reduction under the stated C_3 structural gates). *Let D_3 be a real symmetric linear correction on the four-dimensional field space. Suppose that it leaves the charged C_2 branch unchanged and preserves the photon-null condition,*

$$D_3 W^1 = D_3 W^2 = D_3 A = 0, \quad (126)$$

and suppose that its inherited scale is ρ_{PR} . Then there is one and only one dimensionless scalar $k \in \mathbb{R}$ such that

$$D_3(k) = \rho_{\text{PR}} k P_Z. \quad (127)$$

The structural gates reduce every admissible tensor to this one-coordinate normal form; the normalized-response conditions below determine that coordinate.

Proof. The three vectors in Eq. (126) span a subspace K contained in $\ker D_3$. Symmetry gives $\langle y, D_3 x \rangle = \langle D_3 y, x \rangle = 0$ for every $y \in K$, so $\text{im } D_3 \subseteq K^\perp = \text{span}\{Z\}$. Hence $D_3 = d P_Z$ for a unique $d \in \mathbb{R}$. Since $\rho_{\text{PR}} > 0$, there is a unique $k = d/\rho_{\text{PR}}$. Conversely, every operator in Eq. (127) obeys the stated gates. Therefore the gates reduce the tensor problem to one scalar coordinate. The remaining normalized-response gates are imposed in Theorem 5.3, after which Theorem 5.4 fixes that coordinate to $k = 2$. $\square$

The exact factor available from the PR-II charge algebra is the following. Let

$$\mathcal{H}_{\text{ch}} = \text{span}_{\mathbb{C}}\{T^+, T^-\}, \quad q := \text{ad}_Q|_{\mathcal{H}_{\text{ch}}}, \quad (128)$$

where the charged basis is orthonormal and $Q = T_3 + Y/2$. The hypercharge generator commutes with the $SU(2)$ adjoint action on this subspace.

Proposition 5.2 (Exact charged-adjoint trace). *With the PR-II generator normalization,*

$$qT^+ = T^+, \quad qT^- = -T^-, \quad \text{Tr}_{\text{ch}}(q^\dagger q) = 2. \quad (129)$$

This result contains no fitted coefficient and no continuous parameter.

Proof. The ladder relations are $[T_3, T^\pm] = \pm T^\pm$. The commuting hypercharge term does not alter them, so the matrix of q in the ordered basis (T^+, T^-) is $\text{diag}(1, -1)$. Consequently $q^\dagger q = I_2$, and its trace is exactly two. $\square$

The minimal PR response class uses only the inherited scale, the physical projector, and a quadratic functional of the normalized charged-adjoint operator. The functional itself is unique under the structural properties of a spectral response.

Theorem 5.3 (Unique normalized quadratic charge response). *Let $\mathcal{F}$ assign a nonnegative real number to every normal finite-dimensional charge operator and satisfy, for every dimension-compatible unitary U and every $a \in \mathbb{C}$,*

$$\begin{aligned}\mathcal{F}(UqU^\dagger) &= \mathcal{F}(q), & \mathcal{F}(q_1 \oplus q_2) &= \mathcal{F}(q_1) + \mathcal{F}(q_2), \\ \mathcal{F}(aq) &= |a|^2 \mathcal{F}(q), & \mathcal{F}([1]) &= 1.\end{aligned}\tag{130}$$

These conditions respectively impose charged-basis invariance, orthogonal-mode additivity, quadratic homogeneity, and unit-charge normalization. Then, uniquely,

$$\mathcal{F}(q) = \text{Tr}(q^\dagger q).\tag{131}$$

Proof. Because q is normal, a unitary transformation diagonalizes it with eigenvalues λ_i . Basis invariance and direct-sum additivity reduce $\mathcal{F}(q)$ to the sum of its one-dimensional mode values. Homogeneity and unit normalization give $\mathcal{F}([\lambda_i]) = |\lambda_i|^2$, hence

$$\mathcal{F}(q) = \sum_i |\lambda_i|^2 = \text{Tr}(q^\dagger q).\tag{132}$$

Every step is forced, proving uniqueness. $\square$

Define the target-independent fixed-kernel response-equation class before selecting its member:

$$\mathfrak{E}_{C_3} = \{E_{\mathcal{F}}(D; q) := D - \rho_{\text{PR}} \mathcal{F}(q) P_Z = 0 \mid \mathcal{F} \text{ satisfies Eq. (130)}\}.\tag{133}$$

The domain of every member is the complete set of normal finite-dimensional charge operators used in Theorem 5.3; agreement only on the one matrix $q = \text{diag}(1, -1)$ is not equation equivalence. The theorem therefore derives, rather than assumes, the unique neutral response law in this class:

$$\boxed{\text{H}_Z : \quad D_3^{\text{PR}} = \rho_{\text{PR}} \text{Tr}_{\text{ch}}(q^\dagger q) P_Z.}\tag{134}$$

Here H_Z is the conclusion of the four response conditions. It preserves the charged C_2 branch and photon nullity, uses only the inherited scale ρ_{PR} , and assigns unit weight to a single normalized charged mode. It is a fixed, target-independent logical closure principle of PR-III, not a fitted parameter or a new P1-level foundational postulate. A complete background-field Hessian satisfying these same gates is an equivalent microscopic presentation of the already fixed response law.

Here uniqueness is a statement about equations modulo physical equivalence, not about typography. Within the declared C_3 class, two residual equations are operator-equivalent when an allowed normalized field or basis change and an everywhere-invertible residual map relate them on the complete admissible domain. They are physically equivalent when this relation preserves the full neutral tensor, quotient solution space, and every registered observable. Matching one mass value is not sufficient. An alternative formula is a distinct C_3 equation only if it satisfies the fixed-kernel classification, charged-sector preservation, photon nullity, and all four normalized-response conditions while remaining physically inequivalent.

Theorem 5.4 (Canonical-closure uniqueness of the C_3 coefficient). *In the target-independently defined response-equation class $\mathfrak{E}_{C_3}$ of Eq. (133), with the PR-II charge algebra, the inherited scale in Eq. (125), and the structural gates in Proposition 5.1, the only member of the family in Eq. (127) is*

$$k = 2, \quad D_{3,Z}^{\text{PR}} = \frac{2}{N_{\text{gen}}^{\text{PR}}} \left(\epsilon_V^{\text{PR}} \right)^2.\tag{135}$$

No free parameter remains in this declared class.

Proof. Theorem 5.3 gives $\mathcal{F}(q) = \text{Tr}(q^\dagger q)$ for every member of $\mathfrak{E}_{C_3}$. Proposition 5.2 then gives $D_3^{\text{PR}} = 2\rho_{\text{PR}}P_Z$. Comparison with Eq. (127), followed by division by the strictly positive ρ_{PR} , gives $k = 2$. Equivalently,

$$\|D_3(k) - 2\rho_{\text{PR}}P_Z\|_F^2 = \rho_{\text{PR}}^2(k-2)^2, \quad (136)$$

where $\|P_Z\|_F^2 = 1$. This expression vanishes if and only if $k = 2$. Therefore every in-class fixed-kernel equation either has the positive normalized-response defect $\rho_{\text{PR}}^2(k-2)^2$, or has $k = 2$ and equals $2\rho_{\text{PR}}P_Z$. Algebraically different residual maps related by a ledger-preserving invertible transformation on the complete charge-operator domain are equivalent and do not constitute additional equations. $\square$

Corollary 5.5 (Equation-level exhaustion of the canonical C_3 class). *Let E_{Tr} denote the residual map obtained from $\mathcal{F}(q) = \text{Tr}(q^\dagger q)$. Then*

$$\mathfrak{E}_{C_3}/\sim_{\text{op}} = \{[E_{\text{Tr}}]_{\text{op}}\}. \quad (137)$$

Thus there is no second inequivalent C_3 equation in this class, and all members induce the same physical quotient solution space. A broader higher-derivative, nonlocal, additional-field, independent-second-jet, or independent-Wilson-coefficient equation is outside this theorem and is not an in-class alternative.

A complete physical- Z background family may represent $\mathbf{H}_Z$ through the displayed determinant. Within the declared class this presentation is required to reduce to the same canonical residual law; it cannot introduce a second coefficient or equation.

Lemma 5.6 (Second variation of a reference-normalized determinant). *Let z be a normalized physical- Z background coordinate and let*

$$\Gamma(z) = \sum_i \eta_i \text{Tr}'_i \log(H_i(z)H_i(0)^{-1}), \quad (138)$$

where η_i includes the statistics sign and determinant weight. Assume that the primed domains are fixed near $z = 0$, every $H_i(0)$ is invertible on its primed domain, the families are twice differentiable, and the relative operators $A_i(z) := H_i(z)H_i(0)^{-1}$ satisfy $A_i(z) - I$, $A'_i(z)$, and $A''_i(z)$ in the determinant/trace class required to differentiate the displayed trace on a common logarithm branch. Assume cyclicity is valid and the regularization adds no multiplicative anomaly. With

$$G_i = H_i(0)^{-1}, \quad V_i = H'_i(0), \quad W_i = H''_i(0), \quad (139)$$

one has

$$\Gamma''(0) = \sum_i \eta_i \text{Tr}'_i (G_i W_i - G_i V_i G_i V_i). \quad (140)$$

Proof. Under the stated hypotheses, $\frac{d}{dz} \text{Tr}' \log A_i = \text{Tr}'(A_i^{-1} A'_i)$. Since $A_i(0) = I$, $A'_i(0) = V_i G_i$, and $A''_i(0) = W_i G_i$, differentiating once more gives

$$\left. \frac{d^2}{dz^2} \text{Tr}' \log A_i(z) \right|_0 = \text{Tr}'(W_i G_i - V_i G_i V_i G_i). \quad (141)$$

Cyclicity converts this expression to the corresponding summand in Eq. (140). $\square$

Equation (66) supplies the determinant architecture, and the displayed operators instantiate its residual electromagnetic background. A complementary physical- Z determinant realization additionally specifies the complete $H_i(z)$, V_i , and W_i , including the vector-ghost-Goldstone organization, matter contribution, spectral normalization, and same-order operator ledger. Under those data, Eq. (140) provides the microscopic route to Eq. (134). The following broader family characterizes the distinct second-jet comparison domain without introducing a PR parameter or using numerical fitting.

Proposition 5.7 (Broader fixed-kernel completion family). *For each fixed $k \in \mathbb{R}$, let*

$$X := \frac{q^\dagger q}{2}, \quad H_k(z) := \exp(k\rho_{\text{PR}} z^2 X) \quad \text{on } \mathcal{H}_{\text{ch}}. \quad (142)$$

Then $H_k(0) = I$, $H_k(z)$ is positive and analytic for real z , and it preserves the same charge algebra and exact group trace. Nevertheless the corresponding reference-normalized determinant

$$\Gamma_k(z) := \frac{1}{2} \text{Tr}_{\text{ch}} \log H_k(z) = \frac{1}{2} k \rho_{\text{PR}} z^2 \quad (143)$$

satisfies $\Gamma_k''(0) = k\rho_{\text{PR}}$.

Proof. The operator $X = I_2/2$ is positive and has unit trace. Functional calculus makes $H_k(z)$ positive and analytic, including for negative k . Since the logarithm of the positive exponential is its self-adjoint exponent,

$$\Gamma_k(z) = \frac{1}{2} k \rho_{\text{PR}} z^2 \text{Tr}_{\text{ch}} X = \frac{1}{2} k \rho_{\text{PR}} z^2, \quad (144)$$

and two differentiations give the claim. $\square$

Taking every other determinant block to be a z -independent spectator and defining the neutral response extracted from this block by

$$D_3^{(k)} := \Gamma_k''(0) P_Z = k \rho_{\text{PR}} P_Z \quad (145)$$

embeds the entire classified family of Eq. (127) in the displayed reference-normalized determinant architecture.

Thus $k = 1$, $k = 2$, and every other fixed value label different neutral Hessians only after an independent second jet is admitted. The PR-III admissibility rules do not admit that extra datum. These deliberately enlarged abstract completions are therefore outside the PR equation class and do not constitute additional solutions. Inside the declared class, unit-mode normalization fixes $k = 2$.

There is also a distinct physical-basis gate. With $c := \cos \theta_W^{\text{PR}}$, $s := \sin \theta_W^{\text{PR}}$, and the frozen PR-II angle satisfying $0 < s, c < 1$, PR-II uses

$$Z = cW^3 - sB, \quad A = sW^3 + cB, \quad c^2 + s^2 = 1, \quad (146)$$

and hence $W^3 = cZ + sA$. Even if the charged-adjoint trace is embedded along the T_3 gauge direction, the literal tensor $2P_{W^3}$ becomes

$$2P_{W^3} = 2 \begin{pmatrix} c^2 & cs \\ cs & s^2 \end{pmatrix}_{(Z,A)}, \quad (147)$$

not $2P_Z$. The normalized physical Z map and cancellation of the photon and mixed components are therefore independent admissibility gates, verified by the source-level reconstruction below; the group trace is not used as a substitute for those gates.

The source-level physical- Z coupling is reconstructed exactly as an in-class consistency identity.

Proposition 5.8 (Physical- Z connection reconstructed from PR-II). *Let $t_Z := T_3 - s^2 Q$, $g_Z := g/c$, and let $q_Z := \text{ad}_{g_Z t_Z}|_{\text{span}\{T^+, T^-\}}$ be the physical- Z charge operator on the charged weak-vector adjoint. The PR-II neutral connection satisfies*

$$gW_\mu^3 T_3 + g'B_\mu \frac{Y}{2} = eA_\mu Q + g_Z Z_\mu t_Z. \quad (148)$$

On the charged weak-vector adjoint,

$$[g_Z t_Z, T^\pm] = \pm g c T^\pm, \quad \text{Tr}_{\text{ch}}(q_Z^\dagger q_Z) = 2g^2 c^2. \quad (149)$$

Consequently the normalized physical charge operator

$$\hat{q}_Z := (gc)^{-1} q_Z \quad (150)$$

equals the dimensionless $q = \text{ad}_Q|_{\mathcal{H}_{\text{ch}}}$ used in Eq. (128), and $\text{Tr}_{\text{ch}}(\hat{q}_Z^\dagger \hat{q}_Z) = 2$. Thus the bare trace two is obtained only after the physical vertex factor gc has been factored into the background coordinate or spectral kernel.

Proof. Inverting the PR-II rotation gives $W^3 = sA + cZ$ and $B = cA - sZ$. Substitution into the neutral connection gives the photon coefficient $gsT_3 + g'cY/2 = eQ$. Using $g' = gs/c$ and $Y/2 = Q - T_3$, the Z coefficient becomes

$$gcT_3 - g's\frac{Y}{2} = \frac{g}{c}(T_3 - s^2Q) = g_Z t_Z. \quad (151)$$

On the weak-vector adjoint $Y = 0$, hence $Q = T_3$, $t_Z = c^2 T_3$, and $[g_Z t_Z, T^\pm] = \pm g c T^\pm$. Squaring the two eigenvalues gives Eq. (149); division by gc gives $\hat{q}_Z = \text{ad}_{T_3}|_{\mathcal{H}_{\text{ch}}} = q$. $\square$

The same reconstruction shows why the complete matter block cannot be silently omitted. Put $x = s^2$ and sum the squared chiral charges $t_Z = T_3 - xQ$ over one PR-II generation, including color multiplicity. Directly from the PR-II charge ledger,

$$S_{ZZ}^{(1)}(x) := \sum_{f \text{ chiral}} n_c(f) t_Z(f)^2 = 2 - 4x + \frac{16}{3}x^2 = \frac{16}{3}\left(x - \frac{3}{8}\right)^2 + \frac{5}{4} > 0, \quad (152)$$

$$S_{AZ}^{(1)}(x) := \sum_{f \text{ chiral}} n_c(f) Q_f t_Z(f) = 2 - \frac{16}{3}x. \quad (153)$$

At the frozen PR-II value $x = 0.231355763298189$, both expressions are nonzero. These charge traces do not by themselves compute a loop kernel, but they prove that neither disappearance of the matter response nor cancellation of the mixed AZ response follows from representation algebra alone.

The exact boundary of the admissible operator class can now be isolated. An independently selectable physical- Z second jet, or seagull, crosses that boundary.

Theorem 5.9 (Exclusion of independent second-jet data). *Suppose a completion of the frozen determinant ledger fixes $H_i(0)$, $H'_i(0)$, the block weights η_i , and all frozen spectra and charge representations, but leaves $H''_j(0)$ unspecified for one common-domain C^2 family of closed quadratic forms with $H_j^{(0)}(0) \geq mI$ and $H_j^{(0)}(z) \geq aH_j^{(0)}(0)$ near zero for constants $m, a > 0$, and with $\eta_j \neq 0$. Write $H_j(0) := H_j^{(0)}(0)$. Let X be a positive finite-rank operator whose range lies in the common form domain, commuting with the frozen charge action and satisfying $\text{Tr}' X = 1$. For any fixed $\lambda \in \mathbb{R}$, define locally, in quadratic-form sense,*

$$H_j^{(\lambda)}(z) := H_j^{(0)}(z) + \frac{\lambda \rho_{\text{PR}}}{2\eta_j} z^2 (H_j^{(0)}(0))^{1/2} X (H_j^{(0)}(0))^{1/2}, \quad (154)$$

and leave every other block unchanged. Then the deformation preserves the entire zero jet and first jet, is a finite-rank relative-form perturbation that remains positive for sufficiently small $|z|$, and changes the determinant Hessian by

$$\Gamma''_\lambda(0) - \Gamma''_0(0) = \lambda \rho_{\text{PR}}. \quad (155)$$

Consequently an independently selectable second jet supplies an additional datum and lies outside the minimal-response class. With the declared minimal-response rule imposed, no such datum is available and $k = 2$ is the sole admissible value.

Proof. The added term and its first derivative vanish at $z = 0$, whereas its second derivative is

$$\frac{\lambda \rho_{\text{PR}}}{\eta_j} H_j(0)^{1/2} X H_j(0)^{1/2}. \quad (156)$$

In Eq. (140), the first-vertex term is therefore unchanged and the new contribution is

$$\eta_j \text{Tr}' \left[H_j(0)^{-1} \frac{\lambda \rho_{\text{PR}}}{\eta_j} H_j(0)^{1/2} X H_j(0)^{1/2} \right] = \lambda \rho_{\text{PR}} \text{Tr}' X = \lambda \rho_{\text{PR}}. \quad (157)$$

The relative perturbation is the bounded finite-rank form $(\lambda \rho_{\text{PR}} / (2\eta_j)) z^2 X$. The coercive bound remains positive whenever $|\lambda \rho_{\text{PR}} / (2\eta_j)| z^2 \|X\| < a$, so positivity is preserved in a sufficiently small neighborhood of $z = 0$. The finite-rank support also makes the displayed cyclic trace identity ordinary and anomaly-free on the deformed subspace. $\square$

A broader effective-field-theory comparison also shows that electroweak gauge invariance alone does not define the admissible PR class. The parameter λ below labels that comparison family and is not introduced into the PR ledger.

Proposition 5.10 (Broader-EFT same-symmetry neutral operator). *Let Φ be the PR-II $Y = 1$ order doublet, let $(M_{Z,0}^{\text{PR}})^2 := g_Z^2 (v_{\text{EW}}^{\text{PR}})^2 / 4$ denote the tree/reference neutral mass, and define*

$$\mathcal{O}_{HD} := (\Phi^\dagger D_\mu \Phi)^* (\Phi^\dagger D^\mu \Phi), \quad \Delta \mathcal{L}_\lambda := \frac{2\lambda \rho_{\text{PR}}}{(v_{\text{EW}}^{\text{PR}})^2} \mathcal{O}_{HD}, \quad \lambda \geq 0. \quad (158)$$

This operator is local, Lorentz invariant, and invariant under $SU(2)_L \times U(1)_Y$. At the locked vacuum it produces

$$\Delta \mathcal{L}_\lambda|_{\Phi_0} = \frac{1}{2} \lambda \rho_{\text{PR}} (M_{Z,0}^{\text{PR}})^2 Z_\mu Z^\mu, \quad (159)$$

while producing no $W^\pm$ or photon mass term. Thus $\delta M_Z^2 / (M_{Z,0}^{\text{PR}})^2 = \lambda \rho_{\text{PR}}$; under the relative mass-squared normalization used by the displayed D_3 numerical ledger, it realizes the entire positive fixed-kernel family $D_3(\lambda) = \lambda \rho_{\text{PR}} P_Z$ without a diagnostic input.

Proof. Gauge covariance gives $D_\mu \Phi \mapsto U D_\mu \Phi$, so $\Phi^\dagger D_\mu \Phi$ and its modulus square are gauge invariant. For $\Phi_0 = (0, v_{\text{EW}}^{\text{PR}})^T / \sqrt{2}$,

$$\Phi_0^\dagger D_\mu \Phi_0 = \frac{i(v_{\text{EW}}^{\text{PR}})^2}{4} (g W_\mu^3 - g' B_\mu) = \frac{i(v_{\text{EW}}^{\text{PR}})^2 g_Z}{4} Z_\mu. \quad (160)$$

Therefore $\mathcal{O}_{HD}|_{\Phi_0} = (v_{\text{EW}}^{\text{PR}})^4 g_Z^2 Z_\mu Z^\mu / 16$. Using $(M_{Z,0}^{\text{PR}})^2 = g_Z^2 (v_{\text{EW}}^{\text{PR}})^2 / 4$ gives Eq. (159). The vacuum contraction contains neither the charged directions nor the orthogonal photon direction. $\square$

This λ -family is not an alternative PR branch, and λ is not introduced into the PR ledger. It belongs to a deliberately enlarged EFT comparison class that permits an independent dimension-six Wilson coefficient. Away from the locked vacuum it also changes scalar/Goldstone interactions. It therefore proves only that gauge symmetry alone cannot select k . The canonical PR finite-tier class excludes independent C_3 Wilson coefficients and requires the four target-independent normalized response properties; these derive H_Z , and within that class $k = 2$ is unique. Any background-Hessian presentation admitted to this class must satisfy the same response gates and therefore represents the same equation.

Exact status of the neutral C_3 coefficient

- **Unconditional algebraic result:** $\text{Tr}_{\text{ch}}(q^\dagger q) = 2$.
- **Canonical uniqueness result:** the unique normalized quadratic response functional gives one and only one coefficient, $k = 2$, with no fitted parameter.
- **Equation-exhaustion result:** every proposed fixed-kernel equation either lies outside the declared C_3 response-functional class or has the positive defect $\rho_{\text{PR}}^2(k-2)^2 > 0$; every member of the declared class is the same trace residual law on the complete charge-operator domain, modulo allowed operator equivalence.
- **Scope result:** broader families with independent second jets or Wilson coefficients are not fixed by zero/first-jet data and gauge symmetry alone; they lie outside the declared minimal PR response class and are not alternative PR branches.
- **Microscopic-representation result:** a gauge-fixed neutral background Hessian is an admissible presentation only when its finite counterterms and same-order power counting reduce to the same H_Z identity; otherwise it defines a different class.

Accordingly, the numerical C_3 branch below evaluates the unique member of the canonical minimal-response class. It is not selected because it is a better empirical fit. Its fixed value is

$$D_{3,Z}^{\text{PR}} = 0.0000772464862565637081786207896848877 \dots \quad (161)$$

In the canonical minimal-response class, the C_3 self-energies are

$$\Pi_W^{C_3} = \Pi_W^{C_2}, \quad \Pi_Z^{C_3} = \Pi_Z^{C_2} + D_{3,Z}^{\text{PR}}, \quad (162)$$

with

$$\Pi_Z^{C_3} = -0.0035108469317154309784857330325970968 \dots \quad (163)$$

The corresponding canonical PR-III weak ledger is

$$M_W^{\text{PR}} = M_W^{C_3} = 80.3739428518621 \text{ GeV}, \quad (164)$$

$$M_Z^{\text{PR}} = M_Z^{C_3} = 91.1876764310439 \text{ GeV}, \quad (165)$$

$$G_F^{\text{PR}} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}, \quad (166)$$

$$\rho_{\text{EW}}^{\text{PR}} = 1.01072468650035. \quad (167)$$

The charged branch is unchanged from C_2 ; only the neutral channel receives the $D_{3,Z}^{\text{PR}}$ refinement.

5.6 One-Sigma Diagnostics for the Canonical Closure Branch

For the selected post-generation diagnostic comparison,

$$M_W^{\text{diag}} = 80.3692 \pm 0.0133 \text{ GeV}, \quad (168)$$

$$M_Z^{\text{diag}} = 91.1876 \pm 0.0021 \text{ GeV}, \quad (169)$$

$$G_F^{\text{diag}} = (1.1663788 \pm 0.0000007) \times 10^{-5} \text{ GeV}^{-2}. \quad (170)$$

The final residuals are

$$M_W^{\text{PR}} - M_W^{\text{diag}} = +0.0047428518621 \text{ GeV}, \quad (171)$$

$$M_Z^{\text{PR}} - M_Z^{\text{diag}} = +0.0000764310439 \text{ GeV}, \quad (172)$$

$$G_F^{\text{PR}} - G_F^{\text{diag}} = +4.201759949 \times 10^{-12} \text{ GeV}^{-2}, \quad (173)$$

corresponding to

$$\boxed{M_W : +0.3566\sigma, \quad M_Z : +0.0364\sigma, \quad G_F : +0.6003\sigma.} \quad (174)$$

The progression from C_1 to the canonical C_3 closure is summarized in Table 4.

Table 4: Electroweak branches. Diagnostic distances are computed only after each branch is generated. The C_3 row uses the target-independent canonical response law and no diagnostic value is used to select it.

Branch	M_W [GeV]	M_Z [GeV]	M_W/σ	M_Z/σ	G_F/σ
C_1	80.08709656	91.34817215	-21.2108	+76.4629	+0.6003
C_2	80.37394285	91.18414199	+0.3566	-1.6467	+0.6003
C_3	80.37394285	91.18767643	+0.3566	+0.0364	+0.6003

5.7 Electroweak branch statement

The low-energy fine-structure value is first generated by the electromagnetic calculation and then transported to the inherited weak scale. The weak couplings and C_1 masses follow from that transported value and the PR displacement scale. The C_2 formulas and the scale ρ_{PR} contain no diagnostic weak mass. For C_3 , the charged-adjoint trace fixes the exact group factor two, while the canonical response law H_Z uniquely maps that factor to the physical Z kernel in the declared minimal class.

Under this target-independent canonical closure, PR-III obtains

$$\boxed{\begin{aligned} M_W^{\text{PR}} &= 80.3739428518621 \text{ GeV}, \\ M_Z^{\text{PR}} &= 91.1876764310439 \text{ GeV}, \\ G_F^{\text{PR}} &= 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}, \end{aligned}} \quad (175)$$

with all three selected precision diagnostics inside one standard deviation. The diagnostics do not establish or select H_Z . Its justification is instead the unique basis-invariant, additive, quadratically homogeneous, unit-normalized response theorem. The displayed values are therefore a no-target-input finite-tier prediction within the declared canonical closure class. A complete neutral Hessian presentation is equivalent only when it reduces to this same response law and therefore cannot change the no-fit uniqueness result.

6 Radiative Stability of the Neutrino Branch

The electromagnetic and electroweak branches determine the radiative environment in which the PR-II neutrino solution is tested for stability. PR-III preserves the inherited, no-fit neutrino spectrum and evaluates whether the normal-ordering branch remains stable at the PR-III radiative scale, preserves its ordering and unitary flavor structure, and remains compatible with post-generation oscillation and direct-mass diagnostics. The inherited central branch is

$$(m_1^{(0)}, m_2^{(0)}, m_3^{(0)}) = (0, 0.008627283010, 0.05010655367) \text{ eV}, \quad (176)$$

with

$$\sum_i m_i^{(0)} = 0.058733836680 \text{ eV}. \quad (177)$$

The inherited PMNS branch is specified by

$$\sin^2 \theta_{12}^{\text{PR}} = 0.299447410807983, \quad (178)$$

$$\sin^2 \theta_{23}^{\text{PR}} = 0.565008899090616, \quad (179)$$

$$\sin^2 \theta_{13}^{\text{PR}} = 0.022270270124743, \quad (180)$$

$$\delta_{\text{CP}}^{\text{PR}} = 195.646778408^\circ. \quad (181)$$

The corresponding effective Majorana branch is

$$m_{\beta\beta}^{(0)} = 1.507694477 \text{ meV}. \quad (182)$$

These quantities are fixed before comparison with oscillation fits, cosmological mass ceilings, direct beta-decay bounds, or neutrinoless-double-beta constraints.

6.1 PR radiative kernel and protected massless branch

The most general PR-III mass correction considered at this tier is written as

$$\Delta m_i^{\text{PR}} = m_i^{(0)} R_{\nu i}^{\text{PR}} + S_{\nu i}^{\text{PR}}, \quad (183)$$

where the multiplicative term has the form

$$R_{\nu i}^{\text{PR}} = \frac{1}{16\pi^2} \left(C_{W\nu} L_{Wi}^{\text{PR}} + C_{Z\nu} L_{Zi}^{\text{PR}} + C_{M\nu} L_{Mi}^{\text{PR}} \right). \quad (184)$$

The spectral factors L_{Wi}^{PR} , L_{Zi}^{PR} , and L_{Mi}^{PR} are constrained by the already closed electromagnetic and electroweak ledgers and by the PR spectral gap. The declared multiplicative class contains no additive term $S_{\nu i}^{\text{PR}}$; introducing one would add an independent Majorana-boundary structure and define a different equation class.

At the multiplicative order used here,

$$m_1^{(0)} = 0 \implies \Delta m_1^{\text{PR}} = 0 \quad (185)$$

because no additive seed belongs to the declared domain. The massless boundary branch is therefore protected at this order rather than lifted by hand.

6.2 Electroweak stability scale

The dimensionless neutrino stability scale is inherited from the same weak-boundary deficit and three-sheet structure that enter the electromagnetic and electroweak refinements:

$$\epsilon_\nu^{\text{PR}} = \frac{\Delta_{\text{EW}}^{\text{HR}}}{\sqrt{N_{\text{gen}}^{\text{PR}}}}, \quad N_{\text{gen}}^{\text{PR}} = 3. \quad (186)$$

Using

$$\Delta_{\text{EW}}^{\text{HR}} = 0.0186442803066928997189287764284030288 \dots, \quad (187)$$

PR-III obtains

$$\epsilon_\nu^{\text{PR}} = 0.0107642802539159840599930614668459536 \dots, \quad (188)$$

$$\left(\epsilon_\nu^{\text{PR}} \right)^2 = 0.0001158697293848455622679311845273316 \dots \quad (189)$$

At the present tier this quantity defines a conservative multiplicative stability envelope rather than an asserted exact loop correction:

$$\left| \Delta m_i^{\text{PR}} \right| \leq m_i^{(0)} \left(\epsilon_\nu^{\text{PR}} \right)^2, \quad \left| \Delta m_{\beta\beta}^{\text{PR}} \right| \leq m_{\beta\beta}^{(0)} \left(\epsilon_\nu^{\text{PR}} \right)^2. \quad (190)$$

The resulting half-widths are

$$\Delta m_1^{\text{env}} = 0, \quad (191)$$

$$\Delta m_2^{\text{env}} = 9.99640947695176 \times 10^{-7} \text{ eV}, \quad (192)$$

$$\Delta m_3^{\text{env}} = 5.80583281415014 \times 10^{-6} \text{ eV}, \quad (193)$$

$$\Delta m_{\beta\beta}^{\text{env}} = 1.74696151045016 \times 10^{-7} \text{ eV} \quad (194)$$

$$= 1.74696151045016 \times 10^{-4} \text{ meV}. \quad (195)$$

6.3 Mass ordering under the PR envelope

The allowed intervals generated by Eq. (190) are

$$0.00862628336905230 \text{ eV} \leq m_2 \leq 0.00862828265094770 \text{ eV}, \quad (196)$$

$$0.0501007478371858 \text{ eV} \leq m_3 \leq 0.0501123595028142 \text{ eV}. \quad (197)$$

The summed mass remains in the interval

$$0.0587270312062382 \text{ eV} \leq \sum_i m_i \leq 0.0587406421537618 \text{ eV}. \quad (198)$$

Because the intervals do not overlap and $m_1 = 0$ remains protected,

$$m_1 < m_2 < m_3 \quad (199)$$

throughout the PR stability envelope.

The central mass-squared splittings are

$$\Delta m_{21}^{2,\text{PR}} = 7.443001213463466 \times 10^{-5} \text{ eV}^2, \quad (200)$$

$$\Delta m_{31}^{2,\text{PR}} = 2.5106667206845905 \times 10^{-3} \text{ eV}^2, \quad (201)$$

$$\Delta m_{32}^{2,\text{PR}} = 2.4362367085499558 \times 10^{-3} \text{ eV}^2. \quad (202)$$

All three remain positive over the full envelope. The normal-ordering branch is therefore stable under the PR-III multiplicative deformation.

6.4 PMNS unitarity and admissible radiative deformation

The inherited angle branch corresponds to

$$\theta_{12}^{\text{PR}} = 33.1763566434167^\circ, \quad (203)$$

$$\theta_{23}^{\text{PR}} = 48.7353104030270^\circ, \quad (204)$$

$$\theta_{13}^{\text{PR}} = 8.58243805858008^\circ. \quad (205)$$

For the first row of the PMNS matrix,

$$|U_{e1}|^2 = 0.684951093794123, \quad (206)$$

$$|U_{e2}|^2 = 0.292778636081134, \quad (207)$$

$$|U_{e3}|^2 = 0.022270270124743, \quad (208)$$

and

$$|U_{e1}|^2 + |U_{e2}|^2 + |U_{e3}|^2 = 1. \quad (209)$$

PR-III does not insert an empirical shift into the PMNS angles. Every admissible radiative deformation in the declared class preserves unitarity and is therefore restricted to

$$U_{\text{PMNS}}^{\text{phys}} = U_{\text{PMNS}}^{(0)} \exp(A_\nu^{\text{PR}}), \quad (A_\nu^{\text{PR}})^\dagger = -A_\nu^{\text{PR}}. \quad (210)$$

To first order,

$$\Delta U_{\text{PMNS}}^{\text{PR}} = U_{\text{PMNS}}^{(0)} A_\nu^{\text{PR}}, \quad (211)$$

so unitarity is maintained by construction. The PR-III class fixes the frozen unitary branch; a nonzero phase-sensitive A_ν^{PR} would add a higher-order structure and is outside this class.

6.5 Effective beta and Majorana masses

The effective electron-neutrino mass is

$$m_\beta^2 = \sum_i |U_{ei}|^2 m_i^2. \quad (212)$$

Using the generated PR spectrum and PMNS branch gives

$$m_\beta^{\text{PR}} = 0.00881502940999797 \text{ eV} = 8.81502940999797 \text{ meV}. \quad (213)$$

The effective Majorana mass is

$$m_{\beta\beta} = \left| \sum_i U_{ei}^2 m_i \right|, \quad (214)$$

with central PR value from Eq. (182). The radiative stability interval is

$$\boxed{1.50751978084895 \text{ meV} \leq m_{\beta\beta} \leq 1.50786917315105 \text{ meV}.} \quad (215)$$

Its half-width is only

$$\Delta m_{\beta\beta}^{\text{env}} = 0.000174696151045016 \text{ meV}, \quad (216)$$

corresponding to a relative envelope of

$$115.869729384846 \text{ ppm}. \quad (217)$$

This is a stability statement, not a claim of experimental detection.

6.6 Post-generation diagnostic comparison

Only after the PR spectrum, ordering, PMNS branch, and stability envelope are fixed are external neutrino references introduced. Using the selected representative normal-ordering comparison values,

$$\Delta m_{21}^{2,\text{diag}} = 7.49 \times 10^{-5} \text{ eV}^2, \quad (218)$$

$$\Delta m_{31}^{2,\text{diag}} = 2.513 \times 10^{-3} \text{ eV}^2, \quad (219)$$

$$\sin^2 \theta_{12}^{\text{diag}} = 0.307, \quad (220)$$

$$\sin^2 \theta_{23}^{\text{diag}} = 0.561, \quad (221)$$

$$\sin^2 \theta_{13}^{\text{diag}} = 0.02215, \quad (222)$$

the relative PR residuals are

$$\frac{\Delta m_{21}^{2,\text{PR}} - \Delta m_{21}^{2,\text{diag}}}{\Delta m_{21}^{2,\text{diag}}} = -0.6275\%, \quad (223)$$

$$\frac{\Delta m_{31}^{2,\text{PR}} - \Delta m_{31}^{2,\text{diag}}}{\Delta m_{31}^{2,\text{diag}}} = -0.09285\%, \quad (224)$$

$$\frac{\sin^2 \theta_{12}^{\text{PR}} - \sin^2 \theta_{12}^{\text{diag}}}{\sin^2 \theta_{12}^{\text{diag}}} = -2.4601\%, \quad (225)$$

$$\frac{\sin^2 \theta_{23}^{\text{PR}} - \sin^2 \theta_{23}^{\text{diag}}}{\sin^2 \theta_{23}^{\text{diag}}} = +0.7146\%, \quad (226)$$

$$\frac{\sin^2 \theta_{13}^{\text{PR}} - \sin^2 \theta_{13}^{\text{diag}}}{\sin^2 \theta_{13}^{\text{diag}}} = +0.5430\%. \quad (227)$$

The branch is therefore diagnostically compatible with the selected oscillation comparison without having used those quantities as generation inputs. For the selected direct and cosmological comparisons [3, 4, 1],

$$\sum_i m_i^{\text{PR}} = 0.058733836680 \text{ eV} < 0.12 \text{ eV}, \quad (228)$$

$$m_\beta^{\text{PR}} = 0.008815029410 \text{ eV} < 0.45 \text{ eV}, \quad (229)$$

$$m_{\beta\beta}^{\text{PR}} = 1.507694477 \text{ meV} < 70 \text{ meV}. \quad (230)$$

The PR mass sum occupies 48.9449% of the selected 0.12 eV ceiling, m_β occupies 1.9589% of the selected direct bound, and $m_{\beta\beta}$ occupies 2.1538% of the selected direct-comparison scale [15]. These are post-generation compatibility statements and should be cited to the corresponding oscillation, cosmology, beta-decay, and neutrinoless-double-beta references in the final bibliography.

Table 5: Generated PR-III neutrino observables and selected post-generation diagnostic status.

Quantity	PR-III value	Status
Ordering	$m_1 < m_2 < m_3$	Stable over PR envelope
$\sum_i m_i$	0.058733836680 eV	Below selected cosmology ceiling
m_β	8.815029410 meV	Below selected direct bound
$m_{\beta\beta}$	1.507694477 meV	Stable and below selected direct scale
Δm_{21}^2	$7.4430012 \times 10^{-5} \text{ eV}^2$	Diagnostic-compatible
Δm_{31}^2	$2.5106667 \times 10^{-3} \text{ eV}^2$	Diagnostic-compatible
PMNS first-row sum	1.0000000000000000	Unitary

6.7 Neutrino closure statement

PR-III establishes that the inherited normal-ordering neutrino branch remains stable in the radiative environment generated by the closed electromagnetic and electroweak sectors. The central spectrum is not retuned. The massless branch remains protected at multiplicative order, the m_2 and m_3 intervals remain separated, all mass-squared splittings remain positive, the PMNS update rule is unitary, and the $m_{\beta\beta}$ envelope is narrow.

The resulting finite-tier statement is

$$\boxed{\begin{aligned} (m_1, m_2, m_3) &= (0, 0.008627283010, 0.05010655367) \text{ eV}, \\ \sum_i m_i &= 0.058733836680 \text{ eV}, \quad m_\beta = 8.815029410 \text{ meV}, \\ m_{\beta\beta} &= 1.507694477 \text{ meV}, \quad \text{normal ordering stable.} \end{aligned}} \quad (231)$$

This is the exact neutrino theorem on the declared multiplicative finite-tier domain: the PR-II branch is radiatively stable and diagnostically compatible without importing neutrino target observables as fitted inputs. An additive Majorana seed, a new phase-sensitive deformation, or an all-orders series adds data not present in this domain and therefore defines a different equation class, not another solution of the PR-III neutrino equation.

7 Strong-Sector Phase-Fiber and One-Loop Running Closure

The final gauge-sector extension of PR-III is the compact color branch. PR-II supplied a strong-coupling anchor and generated quark-sheet hierarchy; PR-III assembles the radiative sign, active-flavor structure, threshold ordering, and compatibility of the anchor with a weak-scale strong-coupling reference into one finite-tier ledger. It performs that construction without importing an external value of α_s , external quark masses, or an external $\Lambda_{\overline{\text{MS}}}$ as generation inputs.

The inherited strong anchor is

$$\alpha_3^{\text{PR}} = 0.117861507469150, \quad (232)$$

or

$$\left(\alpha_3^{\text{PR}}\right)^{-1} = 8.4845342764833368725536701837814243566 \dots \quad (233)$$

This quantity is treated as a compact-boundary PR output. The purpose of the present section is not to fit it, but to determine whether it belongs to a consistent $SU(3)_c$ phase-fiber, threshold, and one-loop running structure.

7.1 Compact Color-Fiber Ledger

The theoretical derivation of the strong-sector dynamics relies on the algebraic classification of the inherited internal color projection, allowing the PR-III architecture to formally anchor the color branch within the topological invariants. We start by defining

$$SU(3)_c, \quad (234)$$

with generators T^a in the fundamental representation and the standard invariant normalization

$$\text{Tr}(T^a T^b) = T_F \delta^{ab}. \quad (235)$$

The PR color ledger fixes

$$N_c = 3, \quad \dim \text{Adj}(SU(3)) = 8, \quad (236)$$

and

$$C_A = 3, \quad C_F = \frac{4}{3}, \quad T_F = \frac{1}{2}. \quad (237)$$

These are group-theoretic identities, not fit parameters. Writing

$$\alpha_3^{\text{PR}} = \frac{(g_3^{\text{PR}})^2}{4\pi}, \quad (238)$$

the PR running object has the form

$$\mu \frac{d\alpha_3^{\text{PR}}}{d\mu} = \beta_3^{\text{PR}}(\alpha_3^{\text{PR}}, P_{\text{color}}, K_3^{\text{PR}}), \quad (239)$$

where P_{color} denotes the compact color projection and K_3^{PR} is the reference-normalized PR spectral kernel.

7.2 Strong Spectral Kernel

To evaluate the scale dependence and radiative transport of the strong coupling, the rigid algebraic identities established above must be dynamically coupled to the internal fluctuation spectrum. This evolution is governed by the native PR strong spectral kernel, which acts as a geometric filter mapping the topological supertrace of the color sector directly into a formal running equation. The one-loop organization is

$$\mu \frac{d\alpha_3^{\text{PR}}}{d\mu} = -\frac{(\alpha_3^{\text{PR}})^2}{2\pi} \beta_0^{\text{PR}} K_3^{\text{PR}} + \mathcal{O}((\alpha_3^{\text{PR}})^3). \quad (240)$$

The spectral supertrace contains the adjoint gluon block, color ghosts, and the active quark color sheets. Color-neutral leptons and electroweak residual backgrounds do not contribute directly to the pure $SU(3)_c$ trace. For a color block a , the admissible discrete kernel class is

$$K_{3a}^{\text{PR}}(\mu_0 \rightarrow \mu) = \sum_n \omega_{3a,n}^{\text{PR}} \log \left[\frac{M_a^2 + \mu^2 + \mu_n^2 (\Lambda_3^{\text{PR}})^2}{M_a^2 + \mu_0^2 + \mu_n^2 (\Lambda_3^{\text{PR}})^2} \right] - R_{3a}^{\text{run}}, \quad (241)$$

where

$$\mu_n^2 = \lambda_n - \lambda_0 \geq (\mu_{\text{min}}^{\text{HR}})^2 = 3.053016666732689. \quad (242)$$

The subtraction R_{3a}^{run} preserves the inherited strong anchor and cannot be selected to force agreement with an external α_s reference. The regulator is therefore tied to the positive PR spectral gap rather than to an arbitrary ultraviolet cutoff. At the present tier,

$$\Lambda_3^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \Xi_3^{\text{PR}} \quad (243)$$

is retained as a symbolic PR matching relation. No numerical value of Ξ_3^{PR} or Λ_3^{PR} is inserted. Consequently, PR-III fixes the group coefficient, threshold ledger, and running sign in its declared domain. A fully integrated function $\alpha_3^{\text{PR}}(\mu)$ requires additional scale data and belongs to a different equation class.

7.3 One-loop color coefficient

To quantify the radiative transport of the strong coupling across the physical spectrum, the geometric kernel must be evaluated against the perturbative color β -function. The leading-order running behavior is determined by the dynamical balance between the gauge-boson self-interactions and the fermionic screening contributions. Extracted directly from the rigid topological invariants established in Eq. (237), the one-loop color coefficient is formally defined as

$$\beta_0^{\text{PR}} = \frac{11}{3} C_A - \frac{4}{3} T_F n_f^{\text{PR}}. \quad (244)$$

Using Eq. (237),

$$\boxed{\beta_0^{\text{PR}} = 11 - \frac{2}{3}n_f^{\text{PR}}.} \quad (245)$$

The critical flavor number is

$$n_f^{\text{crit}} = \frac{33}{2} = 16.5. \quad (246)$$

Hence

$$\beta_0^{\text{PR}} > 0 \quad \text{for} \quad 0 \leq n_f^{\text{PR}} \leq 6. \quad (247)$$

For a positively oriented kernel $K_3^{\text{PR}} > 0$, Eq. (240) therefore maintains a strictly negative sign throughout the full three-generation threshold sequence. This is the exact PR realization of the one-loop asymptotic-freedom condition [7, 11].

The negative one-loop sign proves asymptotic freedom for the declared active-flavor ledger: the coupling decreases toward the ultraviolet, while the one-loop expression grows toward the infrared. This perturbative sign is consistent with the conditions needed for dimensional transmutation, but it does not by itself prove confinement, a mass gap, or the infrared completion. If $n_f^{\text{PR}} \geq 17$, the one-loop sign would reverse and this asymptotic-freedom property would be lost. The three-generation ledger has $n_f^{\text{PR}} = 6 < 16.5$, so it remains on the asymptotically free side of the exact one-loop threshold. The coefficient values are summarized in Table 6.

Table 6: One-loop color coefficient over the PR active-flavor sequence.

n_f^{PR}	0	1	2	3	4	5	6
β_0^{PR}	11	31/3	29/3	9	25/3	23/3	7

At the full three-generation count,

$$\mu \frac{d\alpha_3^{\text{PR}}}{d\mu} \Big|_{n_f=6} = -\frac{7}{2\pi} \left(\alpha_3^{\text{PR}}\right)^2 K_3^{\text{PR}} + \dots \quad (248)$$

Evaluated at the inherited anchor, the leading prefactor is

$$-\frac{7}{2\pi} \left(\alpha_3^{\text{PR}}\right)^2 = -0.0154761223561545\dots, \quad (249)$$

before multiplication by K_3^{PR} .

7.4 PR quark thresholds and active flavors

The dynamical evolution of the strong coupling is intrinsically mediated by the exact sequence of fermion mass thresholds, which sequentially decouple from the radiative transport as the energy scale descends. These decoupling boundaries are not treated as empirical inputs or phenomenological phase transitions. Rather, they are strictly governed by the discrete, geometrically generated quark sheets derived in PR-II. Consequently, the complete active-flavor structure is constructed entirely by the following generated mass spectrum:

$$m_u^{\text{PR}} = 0.002146462595 \text{ GeV}, \quad (250)$$

$$m_d^{\text{PR}} = 0.004761039493 \text{ GeV}, \quad (251)$$

$$m_s^{\text{PR}} = 0.094028510538 \text{ GeV}, \quad (252)$$

$$m_c^{\text{PR}} = 1.274964814257465 \text{ GeV}, \quad (253)$$

$$m_b^{\text{PR}} = 4.1664504826753515 \text{ GeV}, \quad (254)$$

$$m_t^{\text{PR}} = 172.8139732666624 \text{ GeV}, \quad (255)$$

which obey

$$m_u^{\text{PR}} < m_d^{\text{PR}} < m_s^{\text{PR}} < m_c^{\text{PR}} < m_b^{\text{PR}} < m_t^{\text{PR}}. \quad (256)$$

These values are not imported from an external quark-mass table. At the present tier the active-flavor count is represented by the sharp-threshold rule

$$n_f^{\text{PR}}(\mu) = \sum_{q=u,d,s,c,b,t} \Theta(\mu - m_q^{\text{PR}}). \quad (257)$$

The corresponding intervals are shown in Table 7.

Table 7: PR active-flavor intervals and one-loop coefficients.

Scale interval	n_f^{PR}	β_0^{PR}
$0 < \mu < m_u^{\text{PR}}$	0	11
$m_u^{\text{PR}} \leq \mu < m_d^{\text{PR}}$	1	31/3
$m_d^{\text{PR}} \leq \mu < m_s^{\text{PR}}$	2	29/3
$m_s^{\text{PR}} \leq \mu < m_c^{\text{PR}}$	3	9
$m_c^{\text{PR}} \leq \mu < m_b^{\text{PR}}$	4	25/3
$m_b^{\text{PR}} \leq \mu < m_t^{\text{PR}}$	5	23/3
$\mu \geq m_t^{\text{PR}}$	6	7

The electroweak matching scale lies above the generated top threshold:

$$\Lambda_{\text{EW}}^{\text{PR}} - m_t^{\text{PR}} = 0.85662507806592 \text{ GeV} > 0. \quad (258)$$

Therefore

$$n_f^{\text{PR}}(\Lambda_{\text{EW}}^{\text{PR}}) = 6, \quad \beta_0^{\text{PR}}(\Lambda_{\text{EW}}^{\text{PR}}) = 7. \quad (259)$$

Equation (257) is the sharp-threshold rule of the declared one-loop class and fixes its active-flavor ordering and sign. Smooth profiles or higher-loop coefficients alter the admissibility domain and are not alternative solutions of this equation.

7.5 Post-generation strong-anchor comparison

After the PR anchor, group ledger, and threshold structure have been fixed, the result may be compared with a selected weak-scale strong-coupling reference,

$$\alpha_s^{\text{diag}} = 0.1179. \quad (260)$$

The residual is

$$\alpha_3^{\text{PR}} - \alpha_s^{\text{diag}} = -0.000038492530850, \quad (261)$$

or

$$\frac{\alpha_3^{\text{PR}} - \alpha_s^{\text{diag}}}{\alpha_s^{\text{diag}}} = -3.26484570398643 \times 10^{-4}. \quad (262)$$

Equivalently,

$$\boxed{\frac{\alpha_3^{\text{PR}} - \alpha_s^{\text{diag}}}{\alpha_s^{\text{diag}}} = -0.0326484570399\%}. \quad (263)$$

The generated anchor is therefore 326.485 ppm below the selected reference and has ratio

$$\frac{\alpha_3^{\text{PR}}}{\alpha_s^{\text{diag}}} = 0.999673515429601 \dots \quad (264)$$

The diagnostic reference is introduced only at this stage and does not modify Eq. (232) or the threshold ledger [15].

Collectively, these evaluations close the stated one-loop strong-anchor calculation. Given the declared compact color projection, active-flavor ledger, sharp-threshold convention, and positive kernel orientation, the calculation fixes $\beta_0^{\text{PR}} = 7$ at $\Lambda_{\text{EW}}^{\text{PR}}$ and generates $\alpha_3^{\text{PR}} = 0.1178615 \dots$ without importing the diagnostic value as a fit. The result agrees with that diagnostic to sub-0.04%. Different spectral subtractions, an independently supplied Λ_3^{PR} , smooth thresholds, higher loops, confinement, or mass-gap postulates are outside the declared one-loop strong class and do not constitute additional solutions within it.

8 Global Anomaly, Running-Continuity, and No-Fit Consistency

The electromagnetic, electroweak, neutrino, and strong branches have so far been constructed separately from a common inherited projection ledger. The final physical question is whether those branches coexist as one consistent theory. PR-III therefore imposes three global conditions. First, the electroweak and color representation ledger must be free of local and global gauge anomalies. Second, the quantities transferred between sectors must remain continuous under the declared projection maps and matching scales. Third, the numerical agreements reported in the preceding sections must preserve the no-fit separation between generated quantities and diagnostic references. No new correction is introduced in this section, as this would be against the rules of the framework set in PR-I. The analysis is an intersection of the already generated sector branches,

$$\mathcal{L}_{\text{PRIII}} = \mathcal{L}_{\text{em}} \cap \mathcal{L}_{\text{EW}} \cap \mathcal{L}_{\nu} \cap \mathcal{L}_3 \cap \ker \mathcal{A} \cap \mathcal{P}_{\text{no-fit}}, \quad (265)$$

where $\ker \mathcal{A}$ denotes the anomaly-free representation subspace and $\mathcal{P}_{\text{no-fit}}$ denotes the provenance condition that diagnostic observables do not enter the generation maps.

8.1 One-generation Chiral Representation Ledger

All anomaly sums are evaluated using left-handed Weyl fields and the PR-II hypercharge convention [15]

$$Q = T_3 + \frac{Y}{2}. \quad (266)$$

For one generation the representation content is shown in Table 8.

The optional neutral singlet has $Y = 0$ and therefore contributes to none of the anomaly sums below.

Table 8: One-generation left-handed Weyl representation ledger. Charge-conjugated right-handed fields are written as left-handed conjugate fields.

Field	$SU(3)_c \times SU(2)_L$	Y	Multiplicity
Q_L	$(\mathbf{3}, \mathbf{2})$	$1/3$	6
u_R^c	$(\bar{\mathbf{3}}, \mathbf{1})$	$-4/3$	3
d_R^c	$(\bar{\mathbf{3}}, \mathbf{1})$	$2/3$	3
L_L	$(\mathbf{1}, \mathbf{2})$	-1	2
e_R^c	$(\mathbf{1}, \mathbf{1})$	2	1
ν_R^c	$(\mathbf{1}, \mathbf{1})$	0	1, optional

8.2 Local Anomaly Cancellation

The color cubic anomaly cancels because the two fundamental color components of Q_L are balanced by the two conjugate singlet branches:

$$\mathcal{A}[SU(3)_c^3] = 2A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 0, \quad (267)$$

where $A(\bar{\mathbf{3}}) = -A(\mathbf{3})$.

The mixed color–hypercharge anomaly is

$$\begin{aligned} \mathcal{A}[SU(3)_c^2 U(1)_Y] &= 2T_F\left(\frac{1}{3}\right) + T_F\left(-\frac{4}{3}\right) + T_F\left(\frac{2}{3}\right) \\ &= 2\left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)\left(-\frac{4}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{2}{3}\right) \\ &= 0. \end{aligned} \quad (268)$$

The mixed weak–hypercharge anomaly is

$$\begin{aligned} \mathcal{A}[SU(2)_L^2 U(1)_Y] &= 3T_2\left(\frac{1}{3}\right) + T_2(-1) \\ &= 3\left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)(-1) \\ &= 0. \end{aligned} \quad (269)$$

The cubic hypercharge anomaly is

$$\begin{aligned} \mathcal{A}[U(1)_Y^3] &= 6\left(\frac{1}{3}\right)^3 + 3\left(-\frac{4}{3}\right)^3 + 3\left(\frac{2}{3}\right)^3 + 2(-1)^3 + 2^3 \\ &= 0. \end{aligned} \quad (270)$$

Likewise, the mixed gravitational–hypercharge anomaly vanishes:

$$\begin{aligned} \mathcal{A}[\text{grav}^2 U(1)_Y] &= 6\left(\frac{1}{3}\right) + 3\left(-\frac{4}{3}\right) + 3\left(\frac{2}{3}\right) + 2(-1) + 2 \\ &= 0. \end{aligned} \quad (271)$$

Therefore, all local gauge and mixed gravitational anomalies cancel within each generation. Since the PR generation count is

$$N_{\text{gen}}^{\text{PR}} = 3, \quad (272)$$

generation replication multiplies zero anomaly residues by three and does not require additional compensating matter. These cancellations follow from the inherited representation and hypercharge ledger; no external anomaly-canceling fields are inserted [2, 6, 5].

8.3 Global $SU(2)$ and Residual Electromagnetic Consistency

The global $SU(2)$ obstruction depends on the parity of the number of left-handed weak doublets. One PR generation contains three color copies of Q_L and one lepton doublet:

$$N_D^{(1)} = 3 + 1 = 4. \quad (273)$$

For three generations,

$$N_D^{\text{PR}} = N_{\text{gen}}^{\text{PR}} N_D^{(1)} = 3 \times 4 = 12, \quad (274)$$

which is even. The Witten $SU(2)$ anomaly is therefore absent, on a per-generation basis and globally [14].

After projection locking,

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}, \quad (275)$$

the charged matter branches occur in vectorlike left–right pairs:

$$u_L/u_R, \quad d_L/d_R, \quad e_L/e_R, \quad (276)$$

with equal electric charges for the two chiralities, while the neutrino branch is electrically neutral. The residual electromagnetic projection is therefore vectorlike. This result is consistent with the exactly massless photon direction retained by the gauge-null quotient and with the charged-mode inventory used in the alpha calculation.

8.4 Cross-sector Running Continuity

For clarification, we note that the term “continuity” means consistency of the quantities passed from one completed PR sector to the next on the registered finite-tier interfaces. A complete all-scale renormalization-group flow has a different domain and is not an additional solution of this interface theorem. The low-energy electromagnetic result supplies the boundary condition

$$(\alpha_{\text{PR}}(0))^{-1} = 137.0359992075134149588913461016165946357 \dots \quad (277)$$

The electroweak transport gives

$$\left(\alpha_{\text{PR}}^{C_1}(\Lambda_{\text{EW}}^{\text{PR}})\right)^{-1} = (\alpha_{\text{PR}}(0))^{-1} - \Delta_{\text{run}}^{C_1}, \quad (278)$$

with

$$\Delta_{\text{run}}^{C_1} = 8.687829492660492292065934163247, \quad (279)$$

and therefore

$$\left(\alpha_{\text{PR}}^{C_1}(\Lambda_{\text{EW}}^{\text{PR}})\right)^{-1} = 128.3481697148529226668254119383695946357 \dots \quad (280)$$

The inverse coupling decreases in the expected screening direction. The transported coupling then generates the C_3 weak ledger. The self-energy factors satisfy

$$1 + \Pi_W^{C_3} > 0, \quad 1 + \Pi_Z^{C_3} > 0, \quad (281)$$

with

$$\Pi_W^{C_3} = 0.0071761868359439894\dots, \quad \Pi_Z^{C_3} = -0.0035108469317154310\dots \quad (282)$$

We find the final W and Z masses remain real and positive. The same dimensionless bridge appears in the weak-vector and neutrino stability sectors:

$$\epsilon_\nu^{\text{PR}} = \epsilon_V^{\text{PR}} = \frac{\Delta_{\text{EW}}^{\text{HR}}}{\sqrt{N_{\text{gen}}^{\text{PR}}}} = 0.0107642802539159841\dots \quad (283)$$

Therefore, the neutrino envelope is not normalized by a new fitted parameter; it inherits the same compact-boundary suppression scale that generated the electroweak vector split. The electroweak matching scale is also continuous with the strong threshold ledger:

$$\Lambda_{\text{EW}}^{\text{PR}} = 173.67059834472832 \text{ GeV} > m_t^{\text{PR}} = 172.8139732666624 \text{ GeV}. \quad (284)$$

So that

$$n_f^{\text{PR}}(\Lambda_{\text{EW}}^{\text{PR}}) = 6, \quad \beta_0^{\text{PR}}(\Lambda_{\text{EW}}^{\text{PR}}) = 7. \quad (285)$$

The electroweak scale therefore enters the six-flavor strong branch without an inconsistent threshold assignment. The resulting continuity chain may be summarized as

$$\alpha_{\text{PR}}(0) \longrightarrow \alpha_{\text{PR}}(\Lambda_{\text{EW}}^{\text{PR}}) \longrightarrow \{M_W, M_Z, G_F\}_{\text{PR}} \longrightarrow \epsilon_\nu^{\text{PR}}, \quad (286)$$

together with

$$\Lambda_{\text{EW}}^{\text{PR}} \longrightarrow \{n_f^{\text{PR}}, \beta_0^{\text{PR}}\}_{\text{strong}}. \quad (287)$$

Each arrow carries an already generated PR quantity; no new diagnostic reference is introduced at a sector boundary.

8.5 No-fit Provenance Across Sectors

For each sector s , let $\mathcal{I}_s^{\text{PR}}$ denote the inherited or previously generated PR inputs and let $\mathcal{E}_s$ denote the external quantities used for later comparison. The global no-fit condition is

$$\mathcal{E}_s \cap \text{Dom}(\mathcal{C}_s) = \emptyset \quad \text{for every completed sector } s. \quad (288)$$

The concrete separation is summarized in Table 9.

Equation (288) is satisfied by every completed sector. Diagnostic comparisons do not alter generated values, and no external fields are added to repair anomaly sums.

8.6 Cross-sector Precision and Compatibility

The principal generated observables and structural conditions are collected in Table 10. Precision quantities are reported in standard deviations where a selected uncertainty is available; compatibility quantities are reported as relative residuals or bound comparisons.

The first four precision observables lie inside their selected one-sigma diagnostic bands. The neutrino and strong branches satisfy the stated compatibility conditions. The anomaly, continuity, and provenance conditions pass simultaneously.

Table 9: Global no-fit separation. The middle column lists the principal PR generation data; the final column lists quantities reserved for post-generation comparison.

Sector	PR generation data	Excluded diagnostic inputs
Electromagnetic	High-resolution spectral ledger, p_1^{HR} , $q_{\text{bc}}^{\text{HR}}$, c_{bc} , $\Delta_{\text{EW}}^{\text{HR}}$, $N_{\text{gen}}^{\text{PR}}$, $\sin^2 \theta_W^{\text{PR}}$, D_1 , D_2	Empirical or recommended α , manually fitted alpha residuals, weak or strong target values
Electroweak	Generated $\alpha_{\text{PR}}(0)$, compact-boundary data, $v_{\text{EW}}^{\text{PR}}$, $\Lambda_{\text{EW}}^{\text{PR}}$, C_1 , C_2 , and C_3	Measured M_W , M_Z , G_F , measured weak angle, fitted κ_W, κ_Z , external oblique parameters
Neutrino	Frozen PR-II spectrum and PMNS branch, PR electroweak stability scale, multiplicative envelope	Measured splittings and PMNS angles, cosmological mass ceilings, beta-decay and $0\nu\beta\beta$ limits
Strong	Frozen α_3^{PR} , $SU(3)_c$ identities, PR quark thresholds, generation count, PR spectral gap	External α_s as a fit input, external quark masses, $\Lambda_{\overline{\text{MS}}}$, hadron or lattice targets
Global structure	Inherited representation ledger, $Q = T_3 + Y/2$, locked sector outputs	External anomaly-canceling fields, empirical charge adjustments, observed CKM or PMNS entries

Table 10: PR-III cross-sector closure summary.

Quantity or condition	PR-III result	Status
α^{-1}	137.0359992075134...	+0.1376 σ
M_W	80.3739428518621 GeV	+0.3566 σ
M_Z	91.1876764310439 GeV	+0.0364 σ
G_F	$1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}$	+0.6003 σ
Δm_{21}^2	$7.4430012 \times 10^{-5} \text{ eV}^2$	−0.6275% relative
Δm_{31}^2	$2.5106667 \times 10^{-3} \text{ eV}^2$	−0.09285% relative
$\sum_i m_i$	0.058733836680 eV	Below selected 0.12 eV ceiling
m_β	8.815029410 meV	Below selected direct bound
$m_{\beta\beta}$	1.507694477 meV	Stable and below selected direct scale
α_3	0.117861507469150	−0.0326485% relative
Local anomalies	All five sums vanish per generation	Pass
Witten anomaly	12 weak doublets	Absent
Residual $U(1)_{\text{em}}$	Charged branches vectorlike	Pass
Running continuity	All declared sector interfaces are consistent	Pass
No-fit provenance	Diagnostic inputs excluded from generation maps	Pass

8.7 Global PR-III Consistency Statement

The global intersection in Eq. (265) is nonempty and contains the canonical PR-III branch, with its neutral C_3 component fixed by the canonical response law H_Z . Its diagnostic and structural status can be summarized as

$$\boxed{\begin{array}{l} \alpha^{-1}, M_W, M_Z, G_F \quad \text{inside selected one-sigma diagnostics,} \\ \text{the } C_3 \text{ neutral value unique in the minimal response class,} \\ \text{neutrino and strong branches diagnostically compatible,} \\ \mathcal{A}_{\text{local}} = 0, \quad \mathcal{A}_{\text{Witten}} = 0, \quad U(1)_{\text{em}} \text{ vectorlike,} \\ \text{cross-sector continuity and no-fit provenance preserved.} \end{array}} \quad (289)$$

This establishes simultaneous diagnostic and structural consistency of the stated branch. The C_3 value is selected by the target-independent closure theorem, not by this diagnostic intersection.

The domain of this result is the registered finite-tier PR-III architecture, not the class of arbitrary all-orders Standard-Model, QCD, or EFT constructions. Complete neutral operators that reduce to H_Z are equivalent presentations. All-orders electromagnetic series, independent Majorana seeds, new phase-sensitive deformations, smooth strong thresholds, higher-loop color running, and nonperturbative modules add structures excluded from the declared domain; they are not unresolved PR-III solutions.

The precise claim established here is finite-tier and class-relative: the stated PR-III branches coexist in one anomaly-free, continuity-preserving, no-target-input ledger whose current precision and compatibility diagnostics are mutually consistent, with the canonical theorem status of the neutral branch explicit.

9 Closed Admissibility Domain and Equation-Architecture Uniqueness

PR-III closes a precisely declared radiative and cross-sector equation class inherited from PR-I and PR-II. The electromagnetic, electroweak, neutrino, strong, anomaly, continuity, and provenance relations are not a menu of fitted alternatives: they form one registered construction whose inputs, operations, normalizations, orders, and equivalence rules are fixed before diagnostic comparison. The central result is

$$\boxed{\text{the completed admissible PR-I–PR-III equation quotient is a singleton.}} \quad (290)$$

Here “singleton” is class-relative. It means that no second inequivalent equation satisfies the same frozen postulates and admissibility rules. It does not assert uniqueness among theories with different postulates, additional fields, higher orders, or independent EFT coefficients, and it is not a claim that an equation with several mathematical roots has only one physical state. Those are different domains, not alternative solutions inside this theorem.

9.1 Frozen Inheritance and Registered Generation Map

The PR-III generation map is the composition

$$\mathcal{G}_{\text{PR-III}} = \mathcal{F}_{\text{III}}(\mathcal{I}_{\text{PR-I}}, \mathcal{I}_{\text{PR-II}}), \quad (291)$$

where $\mathcal{I}_{\text{PR-I}}$ contains the radial spectral and compact-phase boundary data and $\mathcal{I}_{\text{PR-II}}$ contains the electroweak, generation, flavor, neutrino, quark-threshold, and strong-anchor branches. The following inherited quantities are frozen coordinates of this domain:

$$q_{bc}^{\text{HR}}, \quad c_{bc}, \quad p_1^{\text{HR}}, \quad \sin^2 \theta_W^{\text{PR}}, \quad v_{\text{EW}}^{\text{PR}}, \quad \Lambda_{\text{EW}}^{\text{PR}}, \quad N_{\text{gen}}^{\text{PR}}, \quad (292)$$

together with the PR-II neutrino branch, quark-sheet thresholds, and strong anchor. Holding these inherited data fixed is the composition rule of the PR-I–PR-III theorem chain. Every PR-III quantity is generated from that chain without using the corresponding diagnostic target, so the inherited map adds no fitted coefficient and no unregistered freedom.

9.2 Admissibility Gates and Equation Equivalence

For each sector s , let $\mathcal{A}_s^{\text{PRIII}}$ be the set of equations that satisfy all of the following gates:

1. they use only the frozen PR-I/PR-II objects and earlier registered PR-III outputs;
2. they preserve the declared finite tier, spectral kernel, threshold convention, field content, and normalization;
3. they introduce no fitted continuous coefficient and no diagnostic observable into the generation map;
4. they preserve the applicable gauge-null, positivity, unitarity, anomaly, and cross-sector continuity identities; and
5. they satisfy the sector-specific minimality rule, including the normalized additive quadratic response law for C_3 .

The finite-tier qualification is therefore an intrinsic defining condition of $\mathcal{A}_s^{\text{PRIII}}$. Adding a radiative order, field, independent second jet, Wilson coefficient, threshold profile, or new postulate changes the domain and produces an out-of-class theory.

Two admissible equations are equivalent, $E \sim_{\text{PR}} E'$, precisely when normalized basis or field changes and an everywhere-invertible residual map relate them on the complete registered domain while preserving the quotient solution set and all ledger observables. Algebraically different notation is therefore not a second equation.

Theorem 9.1 (Closed PR-I–PR-III equation architecture). *Let $\mathfrak{E}_{\text{PRIII}}$ be the product of the completed sector classes $\mathcal{A}_s^{\text{PRIII}}$, subject to the common anomaly, continuity, and no-fit gates above. With the PR-I/PR-II inheritance fixed and the canonical closure rules used in this manuscript and proved selector-by-selector in PR-IV,*

$$\mathfrak{E}_{\text{PRIII}} / \sim_{\text{PR}} = \{[E_{\text{PRIII}}]_{\text{PR}}\}. \quad (293)$$

Hence no inequivalent in-class equation and no free continuous parameter remain.

Proof. The PR-IV axiom-by-axiom ledger reduces every completed selector to its registered normal form. A proposed change either violates at least one gate, has a strictly nonzero registered defect, or is connected to the canonical residual by an allowed invertible presentation. For the neutral selector this reduction is explicit in Theorem 5.4: $\|D_3(k) - 2\rho_{\text{PR}}P_Z\|_F^2 = \rho_{\text{PR}}^2(k-2)^2$, which vanishes exactly at $k=2$. The same ledger-preserving equivalence rule identifies alternate notation and normalized bases. Intersecting the singleton selector quotients with the already proved anomaly, continuity, positivity, unitarity, and no-fit gates leaves exactly the class in Eq. (293). $\square$

9.3 Diagnostic Separation and Zero-parameter Provenance

For every completed sector,

$$\mathcal{E}_s \cap \text{Dom}(\mathcal{C}_s) = \emptyset. \quad (294)$$

Thus empirical references do not select an equation, coefficient, branch, or root. The reported standard-deviation distances and compatibility bounds are evaluations of the already generated output. Their numerical values may track the diagnostic reference convention, but the PR equations and outputs remain fixed. Equation uniqueness follows from the admissibility and equivalence proof above, independently of numerical proximity to a reference value.

9.4 Closed Claim

The PR-I–PR-III chain fixes one class-relative equation architecture. The PR-II charged algebra gives $\text{Tr}_{\text{ch}}(q^\dagger q) = 2$; the unique normalized response law maps it to $D_3 = 2\rho_{\text{PR}}P_Z$; and every other in-class value has the strictly positive defect $\rho_{\text{PR}}^2(k-2)^2$. Across the completed selectors, PR-IV applies the same normal-form, defect, and equivalence tests. Accordingly, every proposed alternative has exactly one of three statuses:

it violates an admissibility gate and is outside the PR class, it has nonzero exact defect and is not a solution, or it is equivalent to the canonical equation and is not distinct.	(295)
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There is no fourth case, no fitted selector, and no unregistered free parameter. Higher-order, nonlocal, additional-field, independent-second-jet, different-threshold, nonperturbative, and EFT constructions are excluded by the domain definition; they are different theory classes, not latent PR-III solutions. This is the closing equation-architecture theorem for the stated PR-I–PR-III guidelines. It neither compares arbitrary foundational theories nor asserts a unique physical state when a canonical equation itself admits more than one root.

10 Conclusion

Projection Relativity III closes the radiative and cross-sector extension of the projection architecture introduced in the PR-I and PR-II preprints. PR-I supplied the master projection field, radial spectral branch, displacement sector, and compact electromagnetic phase. PR-II lifted that boundary geometry into the compact electroweak and flavor fiber. Starting from those fixed structures, PR-III carries the electromagnetic, electroweak, neutrino, strong, anomaly, and running-continuity branches through one no-target-input ledger while preserving the theorem status of each step.

The first result is the promotion of the compact electromagnetic boundary from a tree-level normalization to a radiatively corrected precision quantity. The $D_1 + D_2$ branch gives

$$(\alpha_{\text{PR,phys}})^{-1} = 137.0359992075134149588913461016165946 \dots, \quad (296)$$

which lies at

$$+0.1376\sigma \quad (297)$$

relative to the selected low-energy diagnostic reference. The implemented finite-tier result is evaluated from the high-resolution PR boundary constants, the weak-boundary deficit, the inherited weak angle, and the declared three-sheet scaling. The diagnostic fine-structure value enters only after the branch has been evaluated. Under the completed PR-IV selector rules, an altered D_1/D_2 basis is outside the registered class and an equivalent rewriting is the same equation.

Transporting the implemented electromagnetic boundary condition to the inherited weak displacement scale defines the declared electroweak branch. The compact-boundary C_1 transport and generation-weighted C_2 charged–neutral self-energy relations are followed by the canonical neutral C_3 refinement. The charged–adjoint algebra proves $\text{Tr}_{\text{ch}}(q^\dagger q) = 2$, and the unique normalized quadratic response theorem fixes the

canonical coefficient $k = 2$ in the declared minimal PR class. On that target-independent closure branch, the ledger yields

$$\begin{aligned} M_W^{\text{PR}} &= 80.3739428518621 \text{ GeV}, \\ M_Z^{\text{PR}} &= 91.1876764310439 \text{ GeV}, \\ G_F^{\text{PR}} &= 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}, \end{aligned} \quad (298)$$

with diagnostic positions

$$M_W : +0.3566\sigma, \quad M_Z : +0.0364\sigma, \quad G_F : +0.6003\sigma. \quad (299)$$

A single weak-angle shift gives different post-generation reconstructions of the diagnostic central values, thereby distinguishing the angle-only comparison class without generating the C_1/C_2 relations. Broader determinant families with independent second jets define enlarged operator classes rather than alternative PR branches. A complete neutral-background Hessian is an equivalent microscopic presentation only when it reduces to the already unique normalized response and physical- Z map. The inherited PR-II neutrino branch remains stable in the resulting radiative environment. PR-III obtains

$$(m_1, m_2, m_3) = (0, 0.008627283010, 0.05010655367) \text{ eV}, \quad (300)$$

with

$$\sum_i m_i = 0.058733836680 \text{ eV}, \quad m_\beta = 8.815029410 \text{ meV}, \quad m_{\beta\beta} = 1.507694477 \text{ meV}. \quad (301)$$

The massless branch remains protected at multiplicative order, the normal ordering is preserved throughout the PR stability envelope, all mass-squared splittings remain positive, and the PMNS branch remains unitary. The effective Majorana mass is confined to

$$1.50751978084895 \text{ meV} \leq m_{\beta\beta} \leq 1.50786917315105 \text{ meV}. \quad (302)$$

The neutrino result is therefore a statement of stability and compatibility, not a refit of oscillation or direct mass data. The compact color branch is likewise consistent at the present tier. The $SU(3)_c$ invariants

$$N_c = 3, \quad C_A = 3, \quad C_F = \frac{4}{3}, \quad T_F = \frac{1}{2} \quad (303)$$

combine with the PR-generated quark thresholds to give

$$n_f^{\text{PR}}(\Lambda_{\text{EW}}^{\text{PR}}) = 6, \quad \beta_0^{\text{PR}}(\Lambda_{\text{EW}}^{\text{PR}}) = 7. \quad (304)$$

The inherited strong anchor,

$$\alpha_3^{\text{PR}} = 0.117861507469150, \quad (305)$$

lies

$$-0.0326484570399\% \quad (306)$$

relative to the selected diagnostic reference. PR-III therefore fixes the threshold ordering, active-flavor ledger, and one-loop asymptotic-freedom sign without importing an external $\Lambda_{\overline{\text{MS}}}$, external quark masses, or a fitted strong-coupling target. These sector results coexist in one consistent representation ledger. The local

$$SU(3)_c^3, \quad SU(3)_c^2 U(1)_Y, \quad SU(2)_L^2 U(1)_Y, \quad U(1)_Y^3, \quad \text{grav}^2 U(1)_Y \quad (307)$$

anomalies cancel per generation. The total number of weak doublets is even, so the Witten anomaly is absent, and the residual $U(1)_{\text{em}}$ branch is vectorlike. The low-energy electromagnetic result connects continuously to weak-scale transport; the same compact-boundary suppression scale governs the vector

and neutrino stability branches; and the electroweak matching scale is consistent with the six-flavor strong threshold ledger. The unifying result of PR-III is therefore not any one numerical agreement in isolation. It is the existence of a common projection chain,

$$\begin{array}{l}
 \text{PR-I spectral and compact boundary} \\
 \longrightarrow \text{PR-II electroweak and flavor lift} \\
 \longrightarrow \text{PR-III radiative electromagnetic closure} \\
 \longrightarrow \text{weak-scale precision} \\
 \longrightarrow \text{neutrino and strong-sector stability} \\
 \longrightarrow \text{global anomaly and continuity consistency,}
 \end{array} \tag{308}$$

in which later sectors inherit earlier PR-generated quantities rather than replacing them with external targets. Within the scope defined in this paper, the resulting branch may be summarized as

$$\begin{array}{l}
 \alpha^{-1}, M_W, M_Z, G_F \text{ inside the selected one-sigma diagnostic bands,} \\
 \text{neutral } C_3 \text{ coefficient unique in the canonical response class,} \\
 \text{normal-ordering neutrino branch radiatively stable,} \\
 \text{strong anchor, threshold ordering, and asymptotic-freedom sign consistent,} \\
 \text{local and global anomaly conditions satisfied,} \\
 \text{cross-sector continuity and no-fit provenance preserved.}
 \end{array} \tag{309}$$

The neutral C_3 analysis establishes three logically distinct statements. First, the normalized charged-adjoint invariant is exactly two. Second, the PR-III minimal neutral-response closure uses this invariant to define the unique charged-preserving, photon-null physical- Z correction in the declared finite-tier class, fixing $k = 2$ without a fitted coefficient or new continuous parameter. Every other in-class equation either has a nonzero normalized-response defect or is an equivalent presentation of the same tensor. Third, any complete gauge-fixed physical- Z Hessian admitted to the same class is an equivalent microscopic realization of that closure. The abstract second-jet families and the broader-EFT operator in Proposition 5.10 characterize a deliberately enlarged operator domain and are not alternative PR branches. The present result is therefore a canonical, class-relative no-fit uniqueness theorem; algebraically different but operator-equivalent presentations remain the same physical equation. With the completed PR-IV Global Singleton Theorem, this trichotomy applies across the registered PR-I–PR-III architecture: every proposed alternative is out of class, has a nonzero exact defect, or is equivalent to the canonical equation. Thus the admissible equation quotient is a singleton and contains no fitted or free continuous parameter. Equations using different postulates, added fields, higher orders, independent Wilson coefficients, or nonperturbative modules define different theory classes; they are not additional solutions of the closed PR-I–PR-III system. This class-relative conclusion concerns the uniqueness of the equation architecture and does not assert uniqueness among arbitrary theories or among multiple roots of a canonical equation.

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